

Sigmoid Loss for Language Image Pre-Training

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Abstract

We propose a simple pairwise Sigmoid loss for Language-Image Pre-training (SigLIP). Unlike standard contrastive learning with softmax normalization, the sigmoid loss operates solely on image-text pairs and does not require a global view of the pairwise similarities for normalization. The sigmoid loss simultaneously allows further scaling up the batch size, while also performing better at smaller batch sizes. Combined with Locked-image Tuning, with only four TPUv4 chips, we train a SigLiT model that achieves 84.5% ImageNet zero-shot accuracy in two days. The disentanglement of the batch size from the loss further allows us to study the impact of examples vs pairs and negative to positive ratio. Finally, we push the batch size to the extreme, up to one million, and find that the benefits of growing batch size quickly diminish, with a more reasonable batch size of 32k being sufficient. We release our models at https://github.com/google-research/big_vision and hope our research motivates further explorations in improving the quality and efficiency of language-image pre-training.

1. Introduction

Contrastive pre-training using weak supervision from image-text pairs found on the web is becoming the go-to method for obtaining generic computer vision backbones, slowly replacing pre-training on large labelled multi-class datasets. The high-level idea is to simultaneously learn an aligned representation space for images and texts using paired data. Seminal works CLIP [36] and ALIGN [23] established the viability of this approach at a large scale, and following their success, many large image-text datasets became available privately [59, 13, 21, 49] and publicly [40, 6, 15, 7, 41].

The standard recipe to pre-train such models leverages the image-text contrastive objective. It aligns the image and

Table 1: **SigLiT and SigLIP results.** Sigmoid loss is memory efficient, allows larger batch sizes (BS) that unlocks language image pre-training with a small number of chips. SigLiT model with a frozen public 🌸 B/8 checkpoint [42], trained on the LiT image-text dataset [59] using four TPU-v4 chips for one day, achieves 79.7% 0-shot accuracy on ImageNet. The same setup with a g/14 checkpoint [58] leads to 84.5% accuracy, trained for two days. With a public unlocked 🗄 B/16 image checkpoint [42], trained on the WebLI dataset [13], SigLIP achieves 71.0% 0-shot accuracy using 16 TPU-v4 chips for three days. The last two rows show results with randomly initialized models.

	Image	Text	BS	#TPUv4	Days	INet-0
SigLiT	🌸 B/8	L*	32 k	4	1	79.8
SigLiT	🌸 g/14	L	20 k	4	2	84.5
SigLIP	🗄 B/16	B	16 k	16	3	71.0
SigLIP	B/16	B	32 k	32	2	72.1
SigLIP	B/16	B	32 k	32	5	73.4

* We use a variant of the L model with 12 layers.

text embeddings for matching (positive) image-text pairs while making sure that unrelated (negative) image-text pairs are dissimilar in the embedding space. This is achieved via a batch-level softmax-based contrastive loss, applied twice to normalize the pairwise similarity scores across all images, then all texts. A naive implementation of the softmax is numerically unstable; it is usually stabilized by subtracting the maximum input value before applying the softmax [18], which requires another pass over the full batch.

In this paper, we propose a simpler alternative: the sigmoid loss. It does not require any operation across the full batch and hence greatly simplifies the distributed loss implementation and boosts efficiency. Additionally, it conceptually decouples the batch size from the definition of the task. We compare the proposed sigmoid loss with the standard softmax loss across multiple setups. In particular, we investigate sigmoid-based loss with two promi-

Sigmoid损失用于语言-图像预训练

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摘要

我们提出了一种用于语言-图像预训练的简单成对 Sigmoid损失（SigLIP）。与使用softmax归一化的标准对比学习不同，Sigmoid损失仅作用于图像-文本对，并且不需要对两两相似度进行全局视角的归一化。Sigmoid损失同时允许进一步扩大批量大小，同时在较小的批量大小下也能表现得更好。结合锁定图像微调，仅使用四个TPUv4芯片，我们在两天内训练了一个 SigLiT模型，该模型达到了84.5%的ImageNet零样本准确率。将批量大小与损失解耦，进一步使我们能够研究样本与配对以及负样本与正样本比率的影响。最后，我们将批量大小推到极限，高达一百万，并发现增大批量大小的收益迅速递减，而更合理的32k批量大小就足够了。我们在 https://github.com/google-research/big_vision 上发布我们的模型，并希望我们的研究能激励进一步提升语言-图像预训练质量和效率的探索。

1. 引言

利用网络上图像-文本对的弱监督信号进行对比预训练，正逐渐成为获取通用计算机视觉骨干网络的主流方法，并逐步取代在大规模标注多类数据集上的预训练。其核心思想是通过配对数据，同时学习图像和文本的对齐表示空间。开创性工作CLIP [36]和 ALIGN [23] 确立了大规模下该方法的可行性，在它们成功之后，许多大型图像-文本数据集开始以私有形式 [59, 13, 21, 49] 和公开形式 [40, 6, 15, 7, 41]出现。

预训练此类模型的标准方法利用图像-文本对比目标。它将图像与

*同等贡献

表1: **SigLiT与SigLIP结果。** Sigmoid损失函数具有内存效率，支持更大的批量大小（BS），从而能够使用少量芯片实现语言图像预训练。使用冻结的公共B/8检查点 [42],在LiT图像-文本数据集 [59] 上使用四块TPU-v4芯片训练一天的SigLiT模型，在ImageNet上达到了79.7%的零样本准确率。相同设置下使用g/14检查点 [58]训练两天，准确率达到84.5%。使用公共解锁的B/16图像检查点 [42],在WebLI数据集 [13], 上训练的SigLIP，使用16块TPU-v4芯片训练三天，实现了71.0%的零样本准确率。最后两行展示了随机初始化模型的结果。

	图像	文本	BS	#TPUv4	日	INet-0
SigLiT	🌸 B/8	L*	32 k	4	1	79.8
SigLiT	🌸 g/14	L	20 k	4	2	84.5
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SigLIP	B/16	B	32 k	32	2	72.1
SigLIP	B/16	B	32 k	32	5	73.4

* 我们使用了L模型的变体，包含12个层。

文本嵌入用于匹配（正样本）图像-文本对，同时确保无关（负样本）图像-文本对在嵌入空间中不相似。这是通过一种批次级软最大化对比损失函数实现的，该损失函数被应用两次，先对所有图像、再对所有文本的成对相似度得分进行归一化。软最大化的简单实现在数值上不稳定；通常通过在对所有输入值应用软最大化之前减去最大值来使其稳定 [18],，这需要对完整批次进行另一次遍历。

在本文中，我们提出了一种更简单的替代方案：Sigmoid损失。它不需要在完整批次上进行任何操作，从而大大简化了分布式损失函数的实现并提高了效率。此外，它在概念上将批量大小与任务定义解耦。我们在多种设置下比较了所提出的Sigmoid损失与标准Softmax损失。具体而言，我们研究了基于sigmoid的损失在两种著名的

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nent approaches for image-text learning: CLIP [36] and LiT [59], which we call sigmoid language image pre-training (*SigLIP*) and sigmoid LiT (*SigLiT*), respectively. We find that the sigmoid loss performs significantly better than the softmax loss when the batch size is smaller than 16k. As the train batch size grows, the gap closes. Importantly, the sigmoid loss is symmetric, requires just a single pass, and a typical implementation requires less memory than the softmax loss. This enables successful training of a SigLiT model at a batch size of *one million*. However, we find that the performance saturates with growing batch size, both for softmax and sigmoid. The good news is that a reasonable batch size, i.e. 32 k, is sufficient for image-text pre-training. This conclusion also holds for multilingual SigLIP training on over 100 languages.

In Table 1, we present setups for image-text pre-training that require a moderate amount of TPUv4 chips for training. SigLiT is surprisingly efficient, reaching 79.7% zero-shot accuracy on ImageNet in just a single day on four chips. SigLIP’s more demanding from-scratch training reaches 73.4% zero-shot accuracy in 5 days with 32 TPUv4 chips. This compares favorably to prior works such as FLIP [30] and CLIP [36], which require approximately 5 and 10 days respectively on 256 TPUv3 cores. When fine-tuning a pre-trained vision backbone in SigLIP, denoted as $\hat{\mathcal{G}}$ in Table 1, we found that disabling the weight decay on the pre-trained backbone leads to better results (see Figure 4 for details). We hope our work paves the way for making the nascent language-image pre-training field more accessible.

2. Related Work

Contrastive learning with the sigmoid loss. One prior work proposes a similar sigmoid loss for the task of unsupervised dimensionality reduction [19]; in the scope of contrastive image-text learning, the vast majority of works rely on the softmax-based InfoNCE loss as popularized by [46]. In supervised classification, the sigmoid loss has already been shown to be slightly more effective and robust than the softmax loss [3, 51].

Contrastive language-image pre-training has become popular since CLIP [36] and ALIGN [23] applied softmax contrastive learning [60, 46, 10, 24] to large-scale image-text datasets. Both models perform very well on zero-shot transfer tasks, including classification and retrieval. Follow-up works show that contrastively pre-trained models produce good representations for fine-tuning [53, 16], linear regression [23], object detection [31], semantic segmentation [33] and video tasks [57].

Generative language-image pre-training Besides softmax contrastive pre-training, various alternatives have been proposed. GIT [49], SimVLM [50], and LEMON [21] successfully pre-train models using a generative text decoder

Algorithm 1 Sigmoid loss pseudo-implementation.

```
1 # img_emb      : image model embedding [n, dim]
2 # txt_emb      : text model embedding [n, dim]
3 # t_prime, b   : learnable temperature and bias
4 # n            : mini-batch size
5
6 t = exp(t_prime)
7 zimg = l2_normalize(img_emb)
8 ztxt = l2_normalize(txt_emb)
9 logits = dot(zimg, ztxt.T) * t + b
10 labels = 2 * eye(n) - ones(n) # -1 with diagonal 1
11 l = -sum(log_sigmoid(labels * logits)) / n
```

instead, while CoCa [56] adds such a decoder to the discriminative CLIP/ALIGN setup, thus combining the pros and cons of both approaches into a single very capable model. BLIP [28] further proposes CapFilt which uses the generative decoder to create better captions and the discriminative part of the model to filter pairs. Language-Image pre-training is a very active field and surveys [8] rapidly become outdated.

Efficient language-image pre-training On the other hand, few works have tried making language image pre-training more efficient. LiT [59] and FLIP [30] are notable attempts, the former requires a pre-trained and locked backbone, and the latter sacrifices quality by randomly dropping visual tokens. BASIC [35] and LAION [52] look at scaling batch-size but only go up to 16 k and 160 k respectively, by using many hundreds of chips, and for the former also mixing in a large private classification dataset [35, 55]. The recent Lion optimizer [12] claims to be able to reduce the training cost to reach similar quality.

3. Method

In this section, we first review the widely-used softmax-based contrastive loss. We then introduce the pairwise sigmoid loss and discuss its efficient implementation.

Given a mini-batch $\mathcal{B} = \{(I_1, T_1), (I_2, T_2), \dots\}$ of image-text pairs, the contrastive learning objective encourages embeddings of matching pairs (I_i, T_i) to align with each other, while pushing embeddings of unmatched pairs $(I_i, T_{j \neq i})$ apart. For practical purposes, it is assumed that for all images i , the text associated with a different image j is not related to i , and vice-versa. This assumption is usually noisy and imperfect.

3.1. Softmax loss for language image pre-training

When using the softmax loss to formalize this objective, an image model $f(\cdot)$ and a text model $g(\cdot)$ are trained to

图像-文本学习方法中的应用: CLIP [36] 和LiT [59], , 我们分别称之为sigmoid语言图像预训练 (*SigLIP*) 和 sigmoid LiT (*SigLiT*) 。我们发现, 当批量大小小于 16k时, Sigmoid损失的表现显著优于Softmax损失。随着训练批次大小的增加, 差距逐渐缩小。重要的是, Sigmoid损失是对称的, 仅需单次传递, 且典型实现所需内存少于Softmax损失。这使我们能够成功训练批量大小为一百万的SigLiT模型。然而, 我们发现, 随着批量大小的增加, 无论是Softmax还是Sigmoid, 其性能都会趋于饱和。好消息是, 合理的批量大小(即32k)足以用于图像-文本预训练。这一结论同样适用于超过100种语言的多语言SigLIP训练。

在表1中, 我们展示了需要中等数量TPUv4芯片进行训练的的图像-文本预训练配置。SigLiT 效率惊人, 仅需四块芯片在一天内即可在 ImageNet 上达到 79.7% 的零样本准确率。SigLIP 要求更高的从头训练则需 32 块 TPUv4 芯片, 在 5 天内达到 73.4% 的零样本准确率。这与先前的工作(如 FLIP [30]和 CLIP [36],)相比更具优势, 后者在 256 个 TPUv3 核心上分别需要大约 5 天和 10 天。当在 SigLIP 中对预训练的视觉骨干网络进行微调时(在表 1中表示为该方式), 我们发现, 在预训练骨干网络上禁用权重衰减会带来更好的结果(详情见图 4)。我们希望我们的工作能为让新兴的语言-图像预训练领域更易普及铺平道路。

2. 相关工作

基于 Sigmoid 损失的对比学习。先前有一项工作针对无监督降维任务提出了类似的 Sigmoid 损失 [19]; 在对比性图像-文本学习领域, 绝大多数工作依赖于基于 Softmax 的 InfoNCE 损失, 该损失由 [46] 推广开来。在监督分类中, Sigmoid 损失已被证明比 Softmax 损失略为有效且更稳健 [3, 51]。

对比语言-图像预训练 自CLIP [36] 和ALIGN [23] 将软最大化对比学习 [60, 46, 10, 24] 应用于大规模图像-文本数据集以来已变得流行。两种模型在零样本迁移任务(包括分类和检索)上表现非常出色。后续工作表明, 对比预训练模型为微调 [53, 16], 、线性回归 [23], 、目标检测 [31], 、语义分割 [33]和视频任务 [57] 生成了良好的表征。

生成式语言-图像预训练除了基于Softmax的对比预训练, 还有多种替代方案被提出。GIT [49], 、SimVLM [50], 和LEMON [21] 成功地使用生成式文本解码器对模型进行预训练

算法1Sigmoid损失伪实现。

```
1 # 图像_emb     : 图像模型嵌入 [n, dim]
2 # txt_emb      : 文本模型嵌入 [n, dim]
3 # t_prime, b   : 可学习温度和偏置
4 # n            : 小批量大小
5
6 t =exp(t_prime)
7 zimg =l2_normalize(img_emb)
8 z文本 =l2_归一化(txt_emb)
9 logits =点积(z图像, z文本.T)*t +b
10 标签 = 2 *eye(n) - ones(n) # -1 对角线为1
11 l =-sum(log_S型(标签 * logits))/ n
```

, 而CoCa [56] 将这样的解码器添加到判别式的 CLIP/ALIGN框架中, 从而将两种方法的优点和缺点结合到一个非常强大的模型中。BLIP [28] 进一步提出了 CapFilt, 它利用生成式解码器生成更优质的描述, 并利用模型的判别部分来过滤配对。语言-图像预训练是一个非常活跃的领域, 相关的综述 [8] 很快就会过时。

高效语言-图像预训练 另一方面, 很少有研究尝试让语言图像预训练更高效。LiT [59]和FLIP [30]是值得注意的尝试: 前者需要预训练且锁定的主干网络, 后者则通过随机丢弃视觉令牌来牺牲质量。BASIC [35]和 LAION [52] 关注批次规模的扩展, 但分别仅达到 16k和160k, 这需要使用数百个芯片, 并且前者还混合了大型私有分类数据集 [35, 55]。最近的LION优化器 [12] 声称能降低训练成本以达到相近的质量。

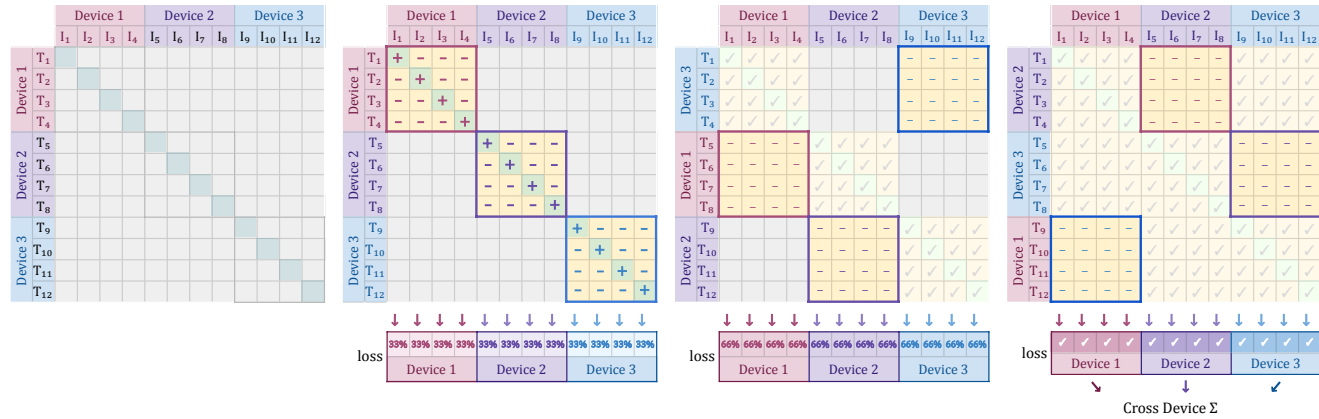
3. 方法

在本节中, 我们首先回顾广泛使用的基于软最大化的对比损失函数。然后, 我们介绍成对Sigmoid损失函数并讨论其高效实现。

给定一批 $\mathcal{B} = \{(I_1, T_1), (I_2, T_2) \dots\}$ 图像-文本对, 对比学习目标鼓励匹配对 (I_i, T_i) 的嵌入彼此对齐, 同时将不匹配对 $(I_i, T_{j \neq i})$ 的嵌入推开。出于实用目的, 假设对于所有图像 i , 与另一图像 j 相关的文本与 i 无关, 反之亦然。这一假设通常带有噪声且不完美。

3.1. 语言图像预训练的Softmax损失函数

当使用Softmax损失函数来形式化这一目标时, 图像模型 $f(\cdot)$ 和文本模型 $g(\cdot)$ 会被训练以



(a) Initially each device holds 4 image and 4 text representations. Each device needs to see the representations from other devices to calculate the full loss. (b) They each compute the component of the loss (highlighted) for their representations, which includes the positives. (c) Texts are swapped across the devices, so device 1 now has $I_{1:4}$ and $T_{5:8}$ etc. The new loss is computed and accumulated with the previous. (d) This repeats till every image & text pair have interacted, e.g. device 1 has the loss of $I_{1:4}$ and $T_{1:12}$. A final cross-device sum brings everything together.

Figure 1: **Efficient loss implementation** demonstrated via a mock setup with 3 devices and a global batch size of 12. There are no all-gathers, and at any point in time only the bright yellow square (size 4×4) is materialized in memory.

minimize the following objective:

$$-\frac{1}{2|\mathcal{B}|} \sum_{i=1}^{|\mathcal{B}|} \left(\log \frac{e^{t\mathbf{x}_i \cdot \mathbf{y}_i}}{\sum_{j=1}^{|\mathcal{B}|} e^{t\mathbf{x}_i \cdot \mathbf{y}_j}} + \log \frac{e^{t\mathbf{x}_i \cdot \mathbf{y}_i}}{\sum_{j=1}^{|\mathcal{B}|} e^{t\mathbf{x}_j \cdot \mathbf{y}_i}} \right)$$

where $\mathbf{x}_i = \frac{f(I_i)}{\|f(I_i)\|_2}$ and $\mathbf{y}_i = \frac{g(T_i)}{\|g(T_i)\|_2}$. In this paper, we adopt the vision transformer architecture [17] for images and the transformer architecture [47] for texts. Note that due to the asymmetry of the softmax loss, the normalization is independently performed two times: across images and across texts [36]. The scalar t is parametrized as $\exp(t')$, where t' is a global freely learnable parameter.

3.2. Sigmoid loss for language image pre-training

Instead of the softmax-based contrastive loss, we propose a simpler alternative that does not require computing global normalization factors. The sigmoid-based loss processes every image-text pair independently, effectively turning the learning problem into the standard binary classification on the dataset of all pair combinations, with a positive labels for the matching pairs (I_i, T_i) and negative labels for all other pairs $(I_i, T_{j \neq i})$. It is defined as follows:

$$-\frac{1}{|\mathcal{B}|} \sum_{i=1}^{|\mathcal{B}|} \sum_{j=1}^{|\mathcal{B}|} \log \frac{1}{1 + e^{z_{ij}(-t\mathbf{x}_i \cdot \mathbf{y}_j + b)}} \quad \mathcal{L}_{ij}$$

where z_{ij} is the label for a given image and text input, which equals 1 if they are paired and -1 otherwise. At initial-

ization, the heavy imbalance coming from the many negatives dominates the loss, leading to large initial optimization steps attempting to correct this bias. To alleviate this, we introduce an additional learnable bias term b similar to the temperature t . We initialize t' and b to $\log 10$ and -10 respectively. This makes sure the training starts roughly close to the prior and does not require massive over-correction. Algorithm 1 presents a pseudocode implementation of the proposed sigmoid loss for language image pre-training.

3.3. Efficient “chunked” implementation

Contrastive training typically utilizes data parallelism. Computing the loss when data is split across D devices necessitates gathering all embeddings [59] with expensive all-gathers and, more importantly, the materialization of a memory-intensive $|\mathcal{B}| \times |\mathcal{B}|$ matrix of pairwise similarities.

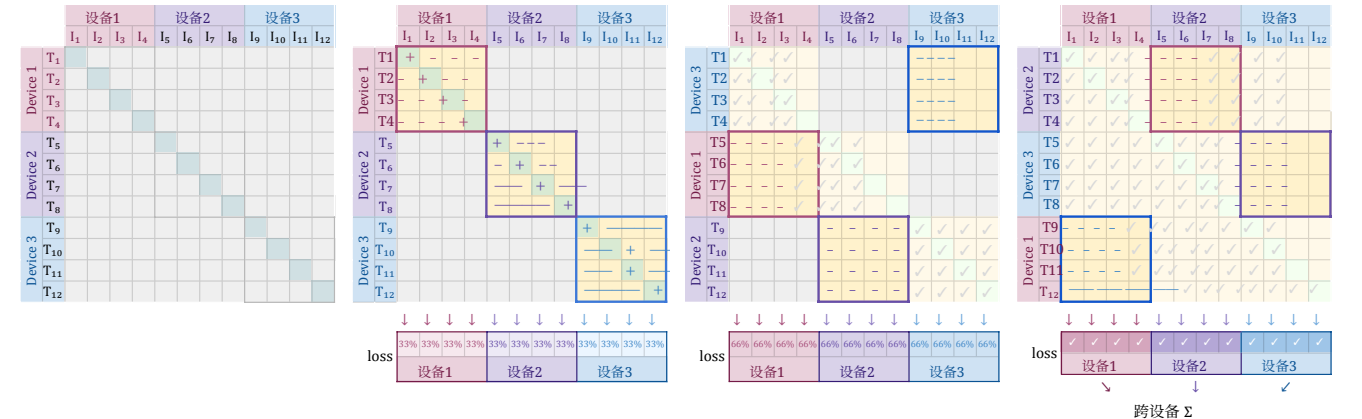
The sigmoid loss, however, is particularly amenable to a memory efficient, fast, and numerically stable implementation that ameliorates both these issues. Denoting the per-device batch size as $b = \frac{|\mathcal{B}|}{D}$, the loss is reformulated as:

$$-\frac{1}{|\mathcal{B}|} \sum_{d_i=1}^D \sum_{d_j=1}^D \sum_{i=bd_i}^{b(d_i+1)} \sum_{j=bd_j}^{b(d_j+1)} \mathcal{L}_{ij}$$

A: $\forall$ device d_i B: swap negs across devices C: per device loss

all local positives negs from next device

This is particularly simple for the sigmoid loss as each pair is an independent term in the loss. Figure 1 illustrates this



(a) 初始状态下，每个设备持有4个图像和4个文本表示。(b) 每个设备为其表示计算损失函数组件（高亮部分），其中包含正样本。每个设备需要查看来自其他设备的表示才能计算完整的损失。(c) 各设备间的文本被交换，因此设备1现在拥有 $I_{1:4}$ 和 $T_{5:8}$ 等。新损失函数被计算并与之前的累计。(d) 此过程重复直至每个图像与文本都对完成交互，例如设备1已获得 $I_{1:4}$ 和 $T_{1:12}$ 的损失函数。最终的跨设备求和将所有结果汇总在一起。

图1：高效的损失实现通过一个包含3个设备和全局批大小为12的模拟设置进行演示。过程中没有全收集操作，且在任一时间点，只有亮黄色方块（大小 4×4 ）在内存中实例化。

最小化以下目标函数：

$$-\frac{1}{2|\mathcal{B}|} \sum_{i=1}^{|\mathcal{B}|} \left(\log \frac{e^{t\mathbf{x}_i \cdot \mathbf{y}_i}}{\sum_{j=1}^{|\mathcal{B}|} e^{t\mathbf{x}_i \cdot \mathbf{y}_j}} + \log \frac{e^{t\mathbf{x}_i \cdot \mathbf{y}_i}}{\sum_{j=1}^{|\mathcal{B}|} e^{t\mathbf{x}_j \cdot \mathbf{y}_i}} \right)$$

其中 $\mathbf{x}_i = \frac{f(I_i)}{\|f(I_i)\|_2}$ 和 $\mathbf{y}_i = \frac{g(T_i)}{\|g(T_i)\|_2}$ 。在本文中，我们采用视觉变换器架构 [17] 用于图像，以及变换器架构 [47] 用于文本。需要注意的是，由于 Softmax 损失的不对称性，归一化操作需独立进行两次：一次跨图像，一次跨文本 [36]。标量 t 被参数化为 $\exp(t')$ ，其中 t' 是一个全局可自由学习的参数。

3.2. Sigmoid损失用于语言-图像预训练

作为基于softmax的对比损失的替代方案，我们提出了一种更简单的方案，无需计算全局归一化因子。基于Sigmoid的损失独立处理每一对图像-文本对，有效地将学习问题转化为标准二分类任务，作用于所有配对组合的数据集，其中匹配对 (I_i, T_i) 为正样本，所有其他对 $(I_i, T_{j \neq i})$ 为负样本。其定义如下：

$$-\frac{1}{|\mathcal{B}|} \sum_{i=1}^{|\mathcal{B}|} \sum_{j=1}^{|\mathcal{B}|} \log \frac{1}{1 + e^{z_{ij}(-t\mathbf{x}_i \cdot \mathbf{y}_j + b)}} \quad \mathcal{L}_{ij}$$

其中 z_{ij} 是针对给定图像和文本输入的标签，如果成对则为 1，否则为 -1 。初始

化时，大量负样本带来的严重不平衡主导了损失函数，导致初始优化步长过大，试图纠正这种偏置。为缓解此问题，我们引入一个额外的可学习偏置项 b ，类似于温度 t 。我们将 t' 和 b 初始化为 $\log 10$ 和 -10 。这确保了训练初始阶段大致接近先前分布，无需大规模过度修正。算法 1 给出了所提出的用于语言图像预训练的Sigmoid损失的伪代码实现。

3.3. 高效的“分块”实现

对比训练通常采用数据并行。当数据分散在 D 各设备上时，计算损失需要借助昂贵的全收集操作来汇聚所有嵌入 [59]，更重要的是，这还需要实例化一个内存密集型的 $|\mathcal{B}| \times |\mathcal{B}|$ 两两相似度矩阵。

然而，Sigmoid损失特别适合实现一种内存高效、快速且数值稳定的方案，从而同时缓解这两个问题。将每台设备的批量大小记为 $b = \frac{|\mathcal{B}|}{D}$ ，损失函数可重新表述为：

$$-\frac{1}{|\mathcal{B}|} \sum_{d_i=1}^D \sum_{d_j=1}^D \sum_{i=bd_i}^{b(d_i+1)} \sum_{j=bd_j}^{b(d_j+1)} \mathcal{L}_{ij}$$

A: $\forall$ 设备 d_i B: 交换负样本跨设备 C: 每设备 loss

全部本地正样本 负样本来自下一个设备

对于Sigmoid损失来说，这尤其简单，因为每一对都是损失中的一个独立项。图1展示了这一点

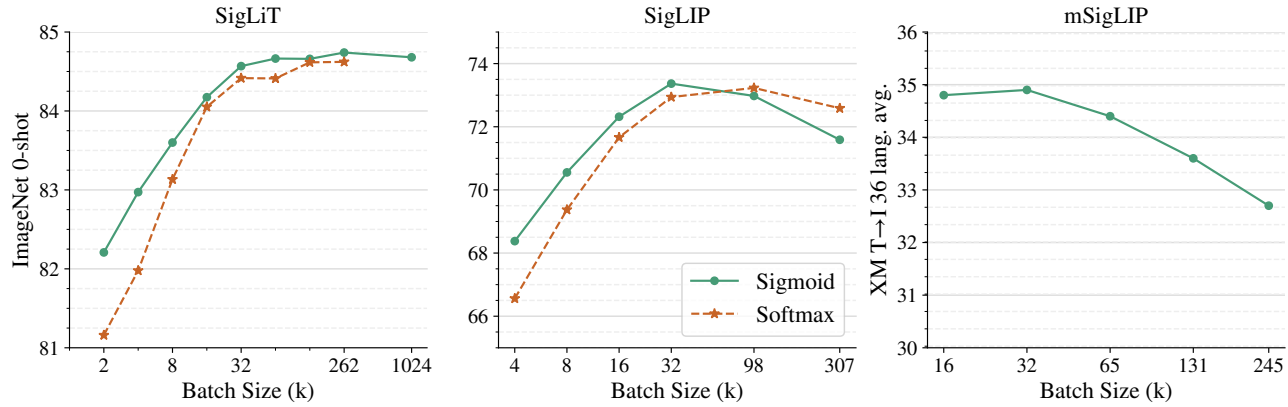


Figure 2: The effect of pre-training batch size. **Left: SigLiT results**, trained for 18B seen examples. Sigmoid loss outperforms the softmax loss significantly with small batch sizes, and performs similarly at larger batch sizes. We successfully trained a SigLiT model with up to *one million* batch size. However, performance for both sigmoid and softmax saturate at around 32 k batch size. **Middle: SigLIP results**, trained for 9B seen examples. Both sigmoid loss and softmax loss saturate at a reasonable batch size, while the peak of the sigmoid loss comes earlier and slightly outperforms the peak of the softmax loss. A very large batch size hurts both losses. **Right: mSigLIP results**, trained for 30B seen examples. With a multilingual setup using over 100 languages, 32 k batch size is surprisingly sufficient and scaling beyond that hurts performance on a 36-language cross-modal retrieval task.

method. In words, we first compute the component of the loss corresponding to the positive pairs, and $b - 1$ negative pairs. We then permute representations across devices, so each device takes negatives from its neighbouring device (next iteration of sum **B**). The loss is then calculated with respect to this chunk (sum **C**). This is done independently in each device, such that each device computes the loss with respect to its local batch b . Losses can then simply be summed across all devices (sum **A**). Individual collective permutes (for sum **B**) are fast (and indeed D collective permutes is typically faster than two all-gathers between D devices), and the memory cost at any given moment is reduced from $|\mathcal{B}|^2$ to b^2 (for sum **C**). Usually b is constant as scaling $|\mathcal{B}|$ is achieved by increasing the number of accelerators. Due to being quadratic with respect to the batch size, the vanilla loss computation rapidly bottlenecks scaling up. This chunked approach enabled training with batch sizes over 1 million on relatively few devices.

4. Results

In this section, we evaluate the proposed SigLiT and SigLIP models across a wide range of batch sizes. We discuss what can be achieved with a small number of accelerator chips, using both SigLiT and SigLIP recipes. We also briefly discuss the impact of batch size on multilingual language image pre-training. We ablate the importance of our large-batch stabilization modification and the introduced learned bias term and present a study on the effect of positive and negative pairs ratio in the sigmoid loss. Lastly,

we explore SigLIP’s data noise robustness.

To validate our models, we report zero-shot transfer results on the ImageNet dataset [14] and zero-shot retrieval results across 36 languages on the XM3600 dataset [44]. We use the ScalingViT-Adafactor optimizer [58] by default for all our experiments.

4.1. SigLiT: Scaling batch size to the limit

Following [59], we use the same precomputed embeddings for the images using a ViT-g vision model, and train a base size text tower from scratch with the same hyperparameters using the LiT image-text dataset [59].

We perform a study over a wide range of batch sizes, from 512 to 1 M , demonstrating the impact of batch size for contrastive learning. Results are presented in Figure 2 (left). When the batch size is smaller than 16 k , sigmoid loss outperforms softmax loss by a large margin. With growing batch sizes, we observe that softmax loss quickly catches up and potentially slightly underperforms sigmoid loss with a large enough batch size. Overall, we recommend using the SigLIP recipe for large batch sizes as well, due to the simplicity, compute savings, and straightforward memory efficient implementation.

There is a consensus that contrastive learning benefits from large batch sizes, while most of the existing studies stop at 64 k batch size [59, 35, 10]. We successfully trained an SigLiT model at one million batch size, to explore the limit of contrastive learning. To our surprise, the performance saturates at 32 k batch size, further scaling up the batch size only gives a minor boost, and the model peaks at

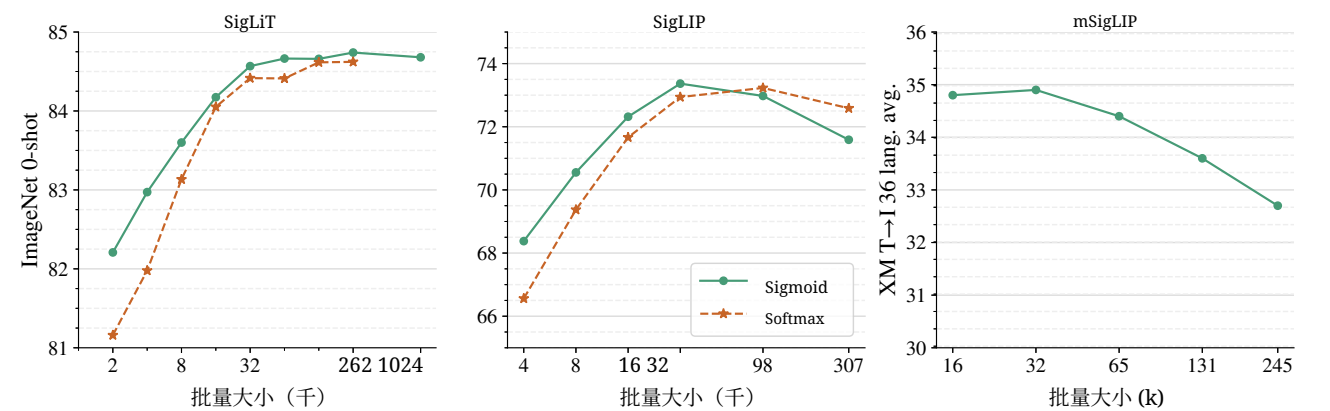


图2: 预训练批量大小的影响。 **左图: SigLiT模型结果**, 训练了180亿个已见样本。在小批量大小下, Sigmoid损失函数显著优于Softmax损失函数, 而在较大批量大小时性能相近。我们成功训练了批量大小高达一百万的SigLiT模型。然而, Sigmoid和Softmax损失函数的性能均在批量大小约32k时趋于饱和。 **中图: SigLIP结果**, 训练了90亿个已见样本。Sigmoid损失函数和Softmax损失函数均在合理的批量大小时达到饱和, 而Sigmoid损失函数的峰值出现更早, 且略优于Softmax损失函数的峰值。非常大的批量大小对两种损失函数都有损害。 **右图: mSigLIP结果**, 训练了300亿个已见样本。在使用超过100种语言的多语言设置下, 32k的批量大小出人意料地足够, 且超过该规模进行扩展反而会损害在36种语言跨模态检索任务上的性能。

方法。简而言之, 我们首先计算与正样本对对应的损失分量, 然后计算负样本对对应的损失分量。然后我们在各设备间对表征进行排列, 使每个设备从相邻设备 (下一次迭代的求和 **B**) 获取负样本。然后针对这一块 (求和 **C**) 计算损失。每个设备独立执行此操作, 使得每个设备针对其本地批次 b 计算损失。接着, 只需将所有设备上的损失相加 (求和 **A**)。单独的集体排列 (用于求和 **B**) 速度很快 (实际上 D 集体排列通常比两次 D 设备间的全收集更快), 并且任意时刻的内存成本从 $|\mathcal{B}|^2$ 降低到 b^2 (用于求和 **C**)。通常, b 是恒定的, 因为扩展 $|\mathcal{B}|$ 是通过增加加速器数量实现的。由于朴素损失计算与批量大小呈二次方关系, 它迅速成为扩展的瓶颈。这种分块方法使得在相对较少的设备上, 能够使用超过100万的批量大小进行训练。

4. 实验结果

在本节中, 我们评估了所提出的 SigLiT 和 SigLIP 模型在多种批量大小下的表现。我们讨论了在使用少量加速器芯片的情况下, 通过 SigLiT 和 SigLIP 配方所能实现的效果。我们还简要探讨了批量大小对多语言语言图像预训练的影响。我们消融研究了大批次稳定化修改的重要性以及引入的学习偏置项, 并呈现了一项关于 Sigmoid 损失中正负样本对比比例影响的研究。最后,

我们探索了 SigLIP 的数据噪声鲁棒性。

为了验证我们的模型, 我们报告了在ImageNet数据集上的零样本迁移结果 [14], 以及在XM3600数据集上跨36种语言的零样本检索结果 [44]。我们默认对所有实验使用ScalingViT-Adafactor优化器 [58]。

4.1. SigLiT: 将批量大小扩展至极限

遵循 [59], 的方法, 我们使用相同的ViT-g视觉模型预计算图像嵌入, 并在LiT图像-文本数据集上从头训练一个基础规模的文本塔, 采用相同的超参数 [59]。

我们针对从512到 1 M 的广泛批量大小范围进行了研究, 展示了批量大小对对比学习的影响。结果如图2 (左) 所示。当批量大小小于 16 k 时, Sigmoid损失以较大优势优于Softmax损失。随着批量大小的增加, 我们观察到 Softmax损失迅速追赶, 并且在批量大小足够大时, 其性能可能略低于Sigmoid损失。总体而言, 鉴于其简洁性、计算节省以及直接高效的内存实现, 我们建议在大量批量大小下也使用SigLIP方案。

学界普遍认为对比学习得益于大批量大小, 而现有研究大多止步于64k批次大小 [59, 35, 10]。我们成功训练了一个一百万批次大小的SigLiT模型, 以探索对比学习的极限。出乎意料的是, 性能在32k批次大小处饱和, 进一步扩大批量大小仅带来微小提升, 模型峰值出现在

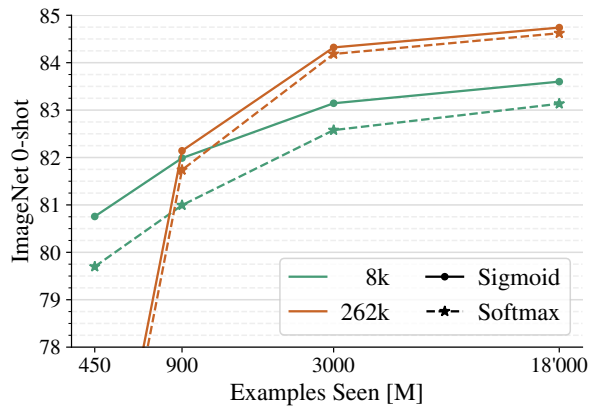


Figure 3: **SigLiT ImageNet 0-shot transfer results with different training durations.** Large batch size results in a big performance boost, but needs a sufficiently long schedule to ramp up, as for short schedules, very large batch size results in a small number of gradient update steps.

256 k batch size. Our best SigLiT with a B -sized text mode achieves 84.7% zero-shot transfer accuracy on ImageNet, while the original LiT paper reports a slightly better 85.2% score with a 10 times larger g -sized text model. Figure 3 presents the impact of training duration for different batch sizes. It demonstrates that large, $262k$ batch size significantly outperforms smaller $8k$ batch size when trained for a sufficiently long time. Note, that for short training durations, large batch size leads to the fewer absolute number of update steps and thus needs more time to ramp up.

4.2. SigLIP: Sigmoid loss is beneficial for language-image pre-training

We pre-train SigLIP models on the WebLI dataset [13], using only English image and text pairs. We use CLIP (WebLI) to denote the CLIP baseline pre-trained on WebLI with the standard softmax loss. We use moderately-sized models: B/16 ViT for image embeddings and B-sized transformer for text embeddings. The input images are resized to 224×224 resolution. The text is tokenized by a 32 k vocabulary sentencepiece tokenizer [27] trained on the English C4 dataset [37], and a maximum of 16 text tokens are kept. Figure 2 middle plot shows SigLIP results, With less than 32 k batch size, SigLIP outperforms CLIP (WebLI) baselines. On the other end of the scale, the memory efficiency of the sigmoid loss enabled much larger batch sizes. For example, with four TPU-v4 chips, we could fit a batch size of 4096 with a Base SigLIP but only 2048 with a corresponding CLIP model. The two advantages together demonstrate significant benefits of the sigmoid loss for language image pre-training with fixed resources, which will be discussed in Section 4.5.

	16 k	32 k	64 k	128 k	240 k
INet-0	71.6	73.2	73.2	73.2	73.1
XM avg	34.8	34.9	34.4	33.6	32.7
XM de	54.7	54.8	55.4	54.3	54.7
XM en	46.5	46.2	46.5	46.6	46.6
XM hi	9.1	8.5	7.9	8.1	7.3
XM ru	50.1	49.9	49.7	48.6	49.3
XM zh	30.7	32.5	32.0	30.6	23.7

Table 2: Multilingual SigLIP results with various batch sizes, pre-trained for 30 billion seen examples. We report zero-shot transfer results on ImageNet (INet-0) and averaged text to image retrieval results across 36 languages on the crossmodal 3600 dataset (XM). The full table on 36 languages can be found in Appendix.

As batch size increases, the gap between the sigmoid and the softmax losses diminish. SigLIP performs best at batch size 32 k, whereas the softmax loss required 98 k for optimal performance and still didn’t outperform the sigmoid based variant. Scaling further, a larger batch size like 307 k hurts both losses.

4.3. mSigLIP: Multi-lingual pre-training

We further scale up the training data by keeping all the *100 languages* from the WebLI dataset [13]. With multilingual data, one usually needs to use a larger international vocabulary. We first verify the impact of two tokenizers: a small multilingual vocabulary with 32 k tokens [37], and a large multilingual vocabulary with 250 k tokens [54]. We train B-sized ViT and text models for 900 M total examples seen, and observe slightly more than 1% improvement when using a larger vocabulary.

However, the token embeddings become huge for very large vocabulary sizes. Following the standard setup, we would need to store a $N \times W$ token embedding lookup table to train the multilingual model, where N is the vocabulary size mentioned above and W is the embedding dimension of the text model. To save memory, we propose to use a “bottlenecked” token embedding. We use $N \times K$ embedding matrix and additional $K \times W$ projection, where the bottleneck K is much smaller than W .

In our experiments, we observed that using a large multilingual vocabulary with a bottleneck can be scaled up as efficiently as using a small multilingual vocabulary. Specifically, by enabling the bottleneck of size $K = 96$ for Base architecture with $W = 768$, we only see about a half percent quality drop on ImageNet zero-shot transfer, compared to using the full 250k vocabulary.

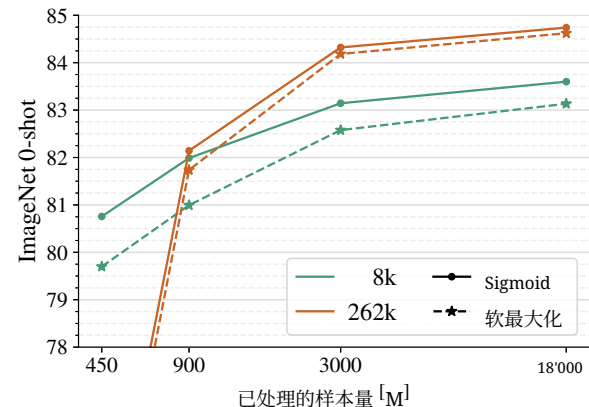


图 3: 不同训练时长下 **SigLiT ImageNet 零样本迁移的结果**。大批量大小能显著提升性能，但需要足够长的训练时长才能充分发挥效果；若训练时长较短，则过大的批量大小会导致梯度更新步数过少。

256k批次大小。我们最佳的SigLiT使用 B 尺寸文本模型在ImageNet上实现了84.7%的零样本迁移准确率，而原始LiT论文中采用大10倍的 g 尺寸文本模型取得了略高的85.2%分数。图3展示了不同批量大小下训练时长的影响。结果表明，当训练时间足够长时，大 $262k$ 批量大小显著优于较小 $8k$ 批量大小。需要注意的是，对于短训练时长，大批量大小会导致绝对更新步数减少，因此需要更多时间才能进入加速阶段。

4.2. SigLIP: Sigmoid损失对语言-图像预训练有益

我们在WebLI数据集上预训练SigLIP模型 [13]，仅使用英文图像和文本对。我们使用CLIP (WebLI) 表示在WebLI上使用标准Softmax损失预训练的CLIP基线模型。我们采用中等规模的模型：B/16 ViT用于图像嵌入，B尺寸变换器用于文本嵌入。输入图像被调整为 224×224 分辨率。文本由32k词汇量的SentencePiece分词器 [27] 在英文C4数据集上训练 [37]，最多保留16个文本令牌。图2中间部分显示SigLIP结果，在小于32k的批量大小下，SigLIP优于CLIP (WebLI) 基线。在规模的另一端，Sigmoid损失的内存效率支持了更大的批量大小。例如，使用四个TPU-v4芯片，基座SigLIP可容纳4096的批量大小，而对应的CLIP模型仅能容纳2048。这两个优势共同表明，Sigmoid损失在固定资源下进行语言图像预训练具有显著优势，这将在第4.5节中讨论。

	16 k	32 k	64 k	128k	240k
INet-0	71.6	73.2	73.2	73.2	73.1
XM 平均	34.8	34.9	34.4	33.6	32.7
XM 德	54.7	54.8	55.4	54.3	54.7
XM 英	46.5	46.2	46.5	46.6	46.6
XM hi	9.1	8.5	7.9	8.1	7.3
XM ru	50.1	49.9	49.7	48.6	49.3
XM zh	30.7	32.5	32.0	30.6	23.7

表2: 在不同批量大小下预训练的多语言SigLIP结果，已处理300亿个可见示例。我们报告了在ImageNet (INet-0) 上的零样本迁移结果，以及在跨模态3600数据集 (XM) 上跨36种语言的文本到图像检索平均结果。包含36种语言的完整表格见附录。

随着批量大小的增加，Sigmoid损失与Softmax损失之间的差距逐渐缩小。SigLIP在批量大小为32k时表现最佳，而Softmax损失需要98k才能达到最佳性能，且仍未能超越基于S型函数的变体。进一步扩展，更大的批量大小（如307k）对两种损失均产生负面影响。

4.3. mSigLIP: 多语言预训练

我们通过保留WebLI数据集中的所有100种语言进一步扩展训练数据 [13]。使用多语言数据时，通常需要采用更大的国际化词汇表。我们首先验证了两种分词器的影响：一个包含32k词元的小型多语言词汇表 [37]，和一个包含250k词元的大型多语言词汇表 [54]。我们训练了B级ViT和文本模型，共处理 900 M 个可见示例，观察到使用更大词汇表时性能提升略高于1%。

然而，对于非常大的词汇量，词元嵌入会变得巨大。按照标准设置，我们需要存储一个 $N \times W$ 词元嵌入查找表来训练多语言模型，其中 N 是上述词汇量， W 是文本模型的嵌入维度。为了节省内存，我们提出使用“瓶颈式”词元嵌入。我们使用 $N \times K$ 嵌入矩阵和额外的 $K \times W$ 投影，其中瓶颈 K 远小于 W 。

在我们的实验中，我们观察到，使用带有瓶颈的大规模多语言词汇表可以像使用小型多语言词汇表一样高效地扩展。具体来说，在 $K = 96$ 大小的瓶颈下，为基础架构启用大小为 $W = 768$ 的瓶颈，与使用完整的 250k 词汇表相比，我们在ImageNet零样本迁移上仅看到约半个百分点的质量下降。

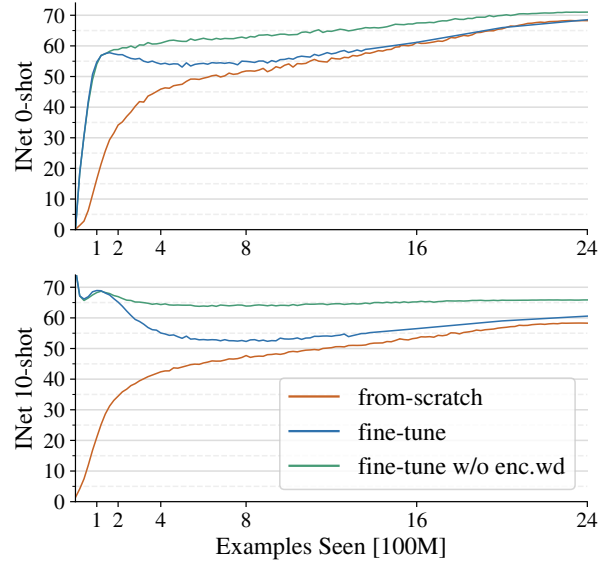


Figure 4: **Top:** SigLIP with pre-trained encoders ramps up quickly. However, only disabling weight decay on the pre-trained encoder weights leads to stable behavior and good ImageNet 0-shot transfer results. **Bottom:** ImageNet 10-shot transfer results, where decaying the pre-trained weights leads to deterioration of the pre-trained model visual representation quality. Disabling weight decay flattens the curve.

With the memory improvements, we train mSigLIP models for various batch sizes, for a total of 30 billion examples seen. Table 2 and Figure 2 (right plot) show the results. We were expecting a large batch size to improve multilingual pre-training, where the model sees more examples from the same language as hard negatives in a single mini-batch. However, we didn’t observe clear improvements with a batch size larger than 32k. A batch size of 32k is sufficient for a multilingual setup as well. On the XM3600 cross-modal retrieval tasks, we found that going beyond 32k batch size leads to worse results on average while on ImageNet zero-shot transfer it stays flat. mSigLIP sets the new state-of-the-art on XM3600 text to image retrieval task, with only a Base size model. Our best result is 34.9%, which is more than 6% higher than the previously reported result 28.5% [13] with a standard LiT model [59] using a much larger four billion ViT-e model. We further scale up mSigLIP training in Section 4.6.

4.4. SigLiT with four TPU-v4 chips

For many practitioners, the important question usually is “what can be trained with a limited amount of resources?” We explore the usage of SigLiT models in this section with only four TPU-v4 chips, as the memory efficient sigmoid loss is suitable for this application scenario.

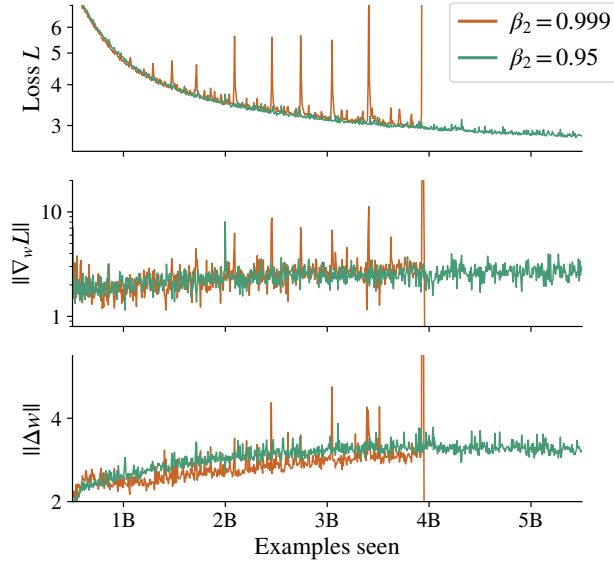


Figure 5: **The effect of Adam and AdaFactor’s β_2 .** As we increase batch-size, we observe more frequent training instability. This instability seen in the loss curves (top) is caused by spikes in gradient norm (middle) leading to large parameter updates (bottom). Decreasing the β_2 momentum stabilizes training. Occasional gradient spikes still happen (see step at 2B), but do not destabilize the training process.

We follow the same setup as in section 4.1. We use the publicly available ViT-AugReg-B/8 [42] model as the frozen (🌀) vision tower, and precompute embeddings to accelerate the training [59]. The text model is a Large Transformer, but with a depth of only 12 layers (instead of 24). It is trained using the LION [12] optimizer with decoupled weight decay 1×10^{-7} , linearly warm-up of learning rate over 6.5k steps up to a peak of 1×10^{-4} , followed by a cosine decay to 0. We train for a total of 65 000 steps with a batch size of 32k – this leads to just under one day of training. Table 1 shows the results when training a model on four chips for one day, achieving 79.7% 0-shot ImageNet classification accuracy; very competitive in this limited resource regime. With a ViT-g/14 [58] model as the vision tower and a Large text tower, we can train at 20k batch size on four chips for 107k steps in under two days. This further pushes the 0-shot ImageNet classification accuracy up to 84.5%.

4.5. SigLIP with a small amount of TPU-v4 chips

It’s resource demanding to train a CLIP model from-scratch in general, with SigLIP it’s possible to fit a larger train batch size with fewer amount of chips. In this section, we explore ways to train SigLIP models efficiently with pre-trained weights. We use pre-trained weights to initialize the image model to accelerate the pre-training, which was orig-

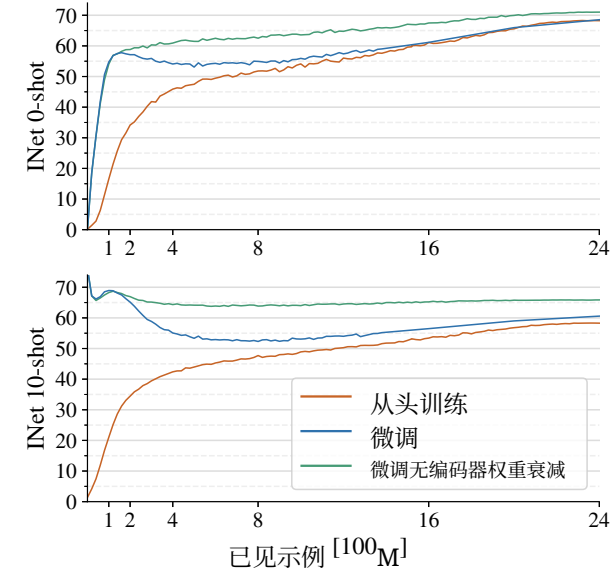


图4: 上图: 使用预训练编码器的SigLIP快速攀升。然而, 仅在预训练编码器权重上禁用权重衰减即可实现稳定行为, 并取得良好的ImageNet零样本迁移结果。下图: ImageNet 10-shot迁移结果, 对预训练权重进行衰减会导致预训练模型视觉表征质量下降。禁用权重衰减可使曲线趋于平缓。

借助内存优化, 我们针对各种批量大小训练了 mSigLIP模型, 总共处理了300亿个样本。表2 和图2 (右侧图) 展示了结果。我们原本预期较大的批量大小能改善多语言预训练, 因为模型会在同一个小批量中从同一语言看到更多作为困难负样本的样本。然而, 我们并未观察到批量大小超过32k时有明显的改进。32k的批量大小对于多语言设置也足够。在XM3600跨模态检索任务上, 我们发现超过32k批量大小会导致平均结果变差, 而在 ImageNet零样本迁移上则保持平稳。mSigLIP在 XM3600文本到图像检索任务上树立了新的最先进水平, 且仅使用基础规模模型。我们的最佳结果是34.9%, 比之前报告的28.5%高出超过6% [13], 后者使用的是标准 LiT模型 [59], 并且采用了更大的40亿参数ViT-e模型。我们将在第4.6节进一步扩展mSigLIP的训练规模。

4.4. 使用四块TPU-v4芯片的SigLiT

对于许多实践者来说, 重要的问题通常是“在有限资源下能训练什么?”在本节中, 我们仅使用四块 TPU-v4芯片探索SigLiT模型的使用, 因为内存高效的 Sigmoid损失适用于这一应用场景。

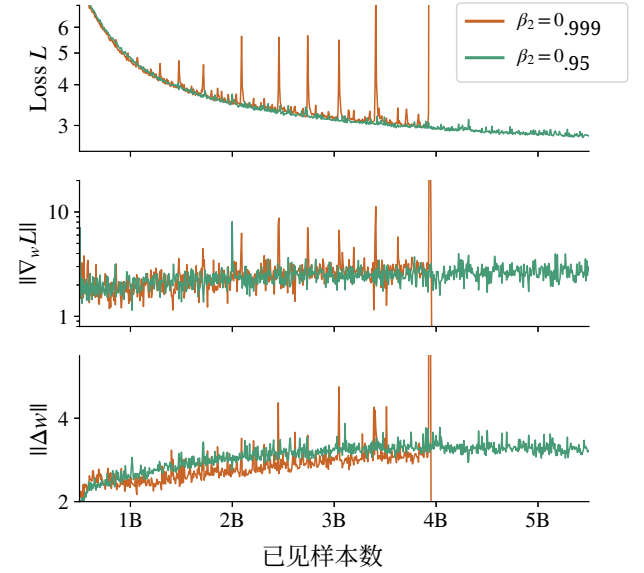


图5: **Adam与AdaFactor的影响 β_2 .** 随着批次大小的增加, 我们观察到训练不稳定性更加频繁。这种在损失曲线 (上图) 中显现的不稳定性, 是由梯度范数 (中图) 的尖峰导致参数更新 (下图) 幅度过大引起的。降低 β_2 动量可稳定训练。偶尔仍会出现梯度尖峰 (例如在20亿步时), 但不会破坏训练过程。

我们采用与第4.1节相同的设置。使用公开的 ViT-AugReg-B/8 [42] 模型作为冻结的(🌀)视觉塔, 并预计算嵌入以加速训练 [59]。文本模型是一个大型 Transformer, 但其深度仅为12层 (而非24层)。它使用LION [12]优化器进行训练, 采用解耦权重衰减 1×10^{-7} , 学习率经过6500步线性预热至峰值 1×10^{-4} , 随后余弦衰减至0。我们总共训练65000步, 批量大小为 32k——这使训练时间不到一天。表1展示了在四块芯片上训练一天的结果, 达到了79.7%的零样本ImageNet分类准确率; 在此有限资源条件下极具竞争力。使用 ViT-g/14 [58] 模型作为视觉塔和大型文本塔, 我们可以在四块芯片上以20k批量大小训练107k步, 耗时不到两天。这进一步将零样本ImageNet分类准确率提升至84.5%。

4.5. 使用少量TPU-v4芯片的SigLIP

从头训练一个CLIP模型通常资源消耗巨大, 而使用SigLIP则能以更少的芯片来容纳更大的训练批次大小。在本节中, 我们将探索利用预训练权重高效训练 SigLIP模型的方法。我们使用预训练权重来初始化图像模型以加速预训练, 这一方法最初

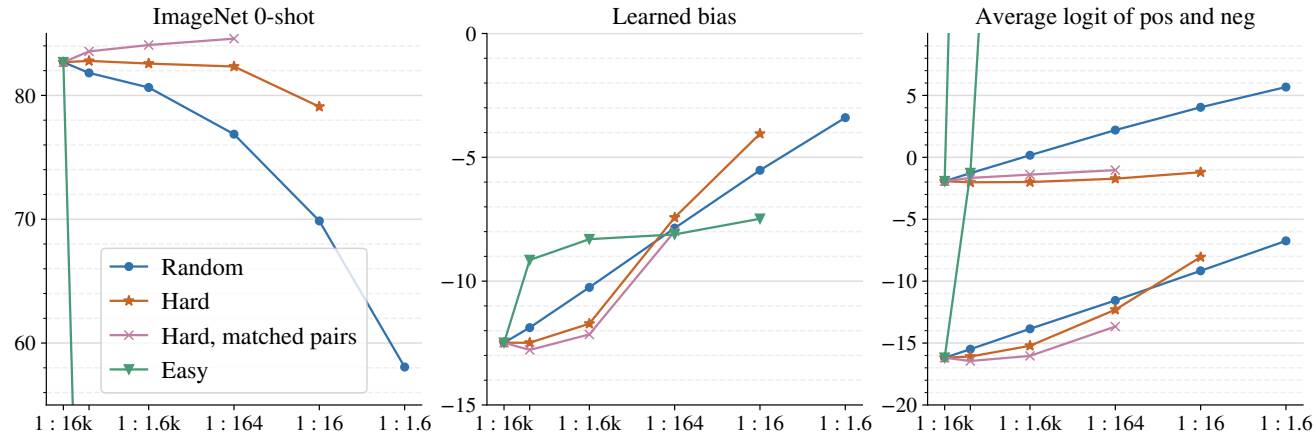


Figure 6: **The effect of batch composition.** We simulate various batch compositions by masking out negatives, either randomly, keeping only the hardest, or the easiest. With no masking, we have 16k negatives for each positive in the batch (1:16k) and the strongest masking we apply (1:1.6) results in almost balanced minibatches. In one setting we *match total pairs* seen by training for significantly longer. We observe ImageNet 0-shot score, the final value of the learned bias, and the average logits of positive and negative pairs. Overall, the imbalance does not seem to be detrimental, but finding an *efficient* way of mining negatives might be beneficial.

inally discussed in [59]. We use the public and unlocked $\square$ ViT-AugReg-B/16 [42] model to initialize our vision tower and fine-tune on the same WebLI English data as used for SigLIP. In all the experiments, we apply a 0.1 learning rate multiplier to the pre-trained image tower to make it suitable for fine-tuning.

Figure 4 presents unlocked $\square$ fine-tuning results alongside from-scratch randomly initialized baselines. We used 16 TPU-v4 chips and train at 16k batch size for 2.4 B examples seen. We found that the fine-tuning setup doesn’t perform well out-of-the-box; this is consistent with prior works [59] where finetuning image models degraded visual representation quality. This is evidenced by ImageNet 10-shot linear classification, where in Figure 4 the fine-tuned setup is barely better than the from-scratch baseline.

We hypothesize that the default weight decay applied to the pre-trained weights reduces their effectiveness. Motivated by the fine-tuning recipe from [17, 58, 25], that uses no weight decay, we also propose disabling weight decay on the pre-trained weights for SigLIP training. Weight decay is therefore only applied to the randomly initialized weights in the text model. This simple modification significantly improved SigLIP results. Figure 4 shows that with our improved recipe, SigLIP reaches 71% 0-shot accuracy on ImageNet, using 16k batch size, trained on 16 chips for three days. We also present from-scratch results in the bottom rows of Table 1: with 32 TPUv4 chips for only two days, SigLIP achieves 72.1% 0-shot accuracy. This presents a significant training cost reduction e.g. compared to CLIP (approx. 2500 TPUv3-days for 72.6%) reported in [30].

4.6. Scaling up SigLIP and mSigLIP

In this section, we scale up SigLIP by “overtraining” the model [45, 1]. We present results in Table 3 using ViT-B, ViT-L or So-400m [1] as the vision encoder, with a text encoder of the same size (B, L and So-400m respectively). Following the recipe described in Section 4.2, we train both models for 40 billion examples seen at batch size 32 k, but use $(256/16)^2 = 256$ image patches and 64 text tokens (instead of 16). To get SigLIP models for different resolutions, we train for 5 billion more examples at the target resolution, with a 100x smaller learning rate and no weight decay. In Table 3, we report zero-shot classification results on ImageNet [14], ObjectNet [2], ImageNet-v2 [39], ImageNet ReaL [3], and zero-shot image-to-text (I→T) retrieval, text-to-image (I→T) retrieval results on MSCOCO [11].

We also scale up the multilingual mSigLIP ViT-B model in the same way. We report image-text retrieval results across 36 languages on the XM3600 benchmark [44]. The scaled-up mSigLIP ViT-B model achieves the state-of-the-art 42.6% *image retrieval recall@1* and 54.1% *text retrieval recall@1* for a Base model. This is slightly outperformed by the Large model in [48] getting 42.96% image retrieval recall@1. Detailed results are provided in Appendix Table 9 and Figure 8, denoted as *32 k.

4.7. Stabilizing large-batch training

As we move to large batch sizes, the language image pre-training using transformers becomes increasingly more unstable, even when using a modestly-sized model (e.g. Base size). The reason for these instabilities is large spikes in the

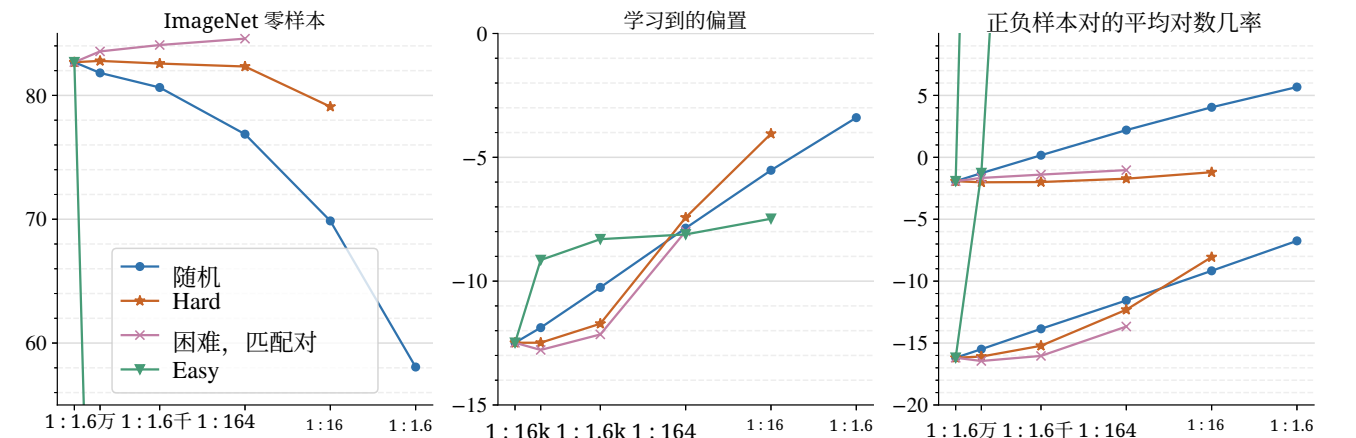


图6: **批次组成的影响。**我们通过掩码负样本来模拟各种批次组成, 掩码方式包括随机掩码、仅保留最困难的负样本或仅保留最简单的负样本。无掩码时, 批次中每个正样本对应16k个负样本 (1:16k); 当应用最强掩码时 (1:1.6), 小批次几乎达到平衡。在一种设置中, 我们通过显著延长训练时间匹配所见的总体本对数量。我们观察了ImageNet零样本分数、学习到的偏置的最终值、以及正负样本对的平均对数几率。总体而言, 不平衡似乎并无害处, 但找到一种高效的负样本挖掘方式可能是有益的。

在 [59]中讨论过。我们采用公开且无锁的 $\square$ ViT-AugReg-B/16 [42] 模型来初始化视觉塔, 并在与 SigLIP相同的WebLI英语数据上进行微调。在所有的实验中, 我们对预训练图像塔应用0.1的学习率乘数, 使其适用于微调。

图4 展示了解锁微调结果 $\square$ 并与从头随机初始化的基线进行了对比。我们使用了16个TPU-v4芯片, 并以16k的批量大小训练, 共处理了24亿个样本。我们发现微调设置开箱即用效果不佳; 这与先前的工作一致 [59], 其中微调图像模型会降低视觉表征质量。这一点可以从ImageNet 10-shot线性分类中看出, 在图4中, 微调后的设置仅比从头训练的基线略好。

我们假设对预训练权重应用的默认权重衰减会降低其有效性。受 [17, 58, 25], 中不使用权重衰减的微调方案启发, 我们也建议在SigLIP训练中禁用对预训练权重的权重衰减。因此, 权重衰减仅应用于文本模型中随机初始化的权重。这一简单修改显著提升了 SigLIP的效果。图4 显示, 采用我们改进的方案, SigLIP在使用 16k 批大小、16个芯片训练三天后, 在 ImageNet上达到了71%的零样本准确率。我们还在表1 下半部分展示了从头训练的结果: 仅用32个TPUv4芯片训练两天, SigLIP就达到了72.1%的零样本准确率。与 [30]中报道的CLIP (约2500 TPUv3天达到72.6%) 相比, 这显著降低了训练成本。

4.6. 扩展 SigLIP 与 mSigLIP

在本节中, 我们通过“过度训练”模型来扩展SigLIP [45, 1]。我们使用ViT-B、ViT-L或So-400m [1]作为视觉编码器, 并搭配相同规模的文本编码器 (分别为B、L和So-400m), 结果展示在表3 中。按照第4.2节描述的方案, 我们以32k的批大小训练两个模型处理400亿个示例, 但使用 $(256/16)^2 = 256$ 个图像块和64个文本令牌 (而非16个)。为了获得不同分辨率的SigLIP模型, 我们在目标分辨率下额外训练50亿个示例, 学习率降低100倍且不使用权重衰减。在表3中, 我们报告了在ImageNet [14], ObjectNet [2], ImageNet-v2 [39], ImageNetReaL [3], 上的零样本分类结果, 以及在MSCOCO [11]上的零样本图像到文本 (I→T) 检索和文本到图像 (I→T) 检索结果。

我们以相同的方式扩展了多语言 mSigLIP ViT-B 模型。我们报告了在 XM3600 基准上跨越 36种语言的图文检索结果 [44]。扩展后的 mSigLIP ViT-B 模型在基础模型中取得了最先进的成绩: 图像检索召回率 @1 为 42.6%, 文本检索召回率@1 为 54.1%。这稍逊于大型模型 [48] 所取得的 42.96% 图像检索召回率 @1。详细结果见附录表9和图8, 标记为 *32k。

4.7. 稳定大批量训练

随着我们转向大批量大小, 使用Transformer进行语言-图像预训练变得越来越不稳定, 即使在使用中等规模模型 (如基础规模) 时也是如此。造成这些不稳定的原因在于

Method	Image Encoder		ImageNet-1k				COCO R@1	
	ViT size	# Patches	Validation	v2	ReaL	ObjectNet	I → T	T → I
CLIP	B	196	68.3	61.9	-	55.3	52.4	33.1
OpenCLIP	B	196	70.2	62.3	-	56.0	59.4	42.3
EVA-CLIP	B	196	74.7	67.0	-	62.3	58.7	42.2
SigLIP	B	196	76.2	69.6	82.8	70.7	64.4	47.2
SigLIP	B	256	76.7	70.0	83.1	71.3	65.1	47.4
SigLIP	B	576	78.6	72.1	84.5	73.8	67.5	49.7
SigLIP	B	1024	79.2	73.0	84.9	74.7	67.6	50.4
CLIP	L	256	75.5	69.0	-	69.9	56.3	36.5
OpenCLIP	L	256	74.0	61.1	-	66.4	62.1	46.1
CLIPA-v2	L	256	79.7	72.8	-	71.1	64.1	46.3
EVA-CLIP	L	256	79.8	72.9	-	75.3	63.7	47.5
SigLIP	L	256	80.5	74.2	85.9	77.9	69.5	51.1
CLIP	L	576	76.6	72.0	-	70.9	57.9	37.1
CLIPA-v2	L	576	80.3	73.5	-	73.1	65.5	47.2
EVA-CLIP	L	576	80.4	73.8	-	78.4	64.1	47.9
SigLIP	L	576	82.1	75.9	87.0	81.0	70.6	52.7
OpenCLIP	G (2B)	256	80.1	73.6	-	73.0	67.3	51.4
CLIPA-v2	H (630M)	576	81.8	75.6	-	77.4	67.2	49.2
EVA-CLIP	E (5B)	256	82.0	75.7	-	79.6	68.8	51.1
SigLIP	SO (400M)	729	83.2	77.2	87.5	82.9	70.2	52.0

Table 3: **Comparison with other publicly released models.** Our SigLIP models outperform all prior models, e.g. OpenCLIP [22] and CLIP [36], by a significant margin on both zero-shot classification and retrieval tasks. Compared to the concurrent EVA-CLIP [43] and CLIPA-v2 [29], our SigLIP-L performs better across the board, in both the low and high resolution cases. Especially noteworthy is the Shape-Optimized 400M parameter ViT [1] architecture, which outperforms all significantly larger models. We publicly release our models: https://github.com/google-research/big_vision.

gradient norms, which translate to large-magnitude changes in the weights that may destabilize the training process, see Figure 5. We observe that reducing β_2 in Adam and AdaFactor from its default 0.999 to 0.95 (which was suggested in [20, 9]) is enough to stabilize the training. Intuitively, this allows recovering from gradient spikes quicker. We opt for setting $\beta_2 = 0.95$ for all our experiments.

4.8. Negative ratio in sigmoid loss

One question which arises when shifting the perspective from the softmax’s “pick the right class” view to the sigmoid’s “rate this pair” view, is the imbalance in positive versus negative pairs. For a batch size $|\mathcal{B}|$, the batch contains $|\mathcal{B}|$ positive pairs, but $|\mathcal{B}|^2 - |\mathcal{B}|$ negative examples. In the modest batch-size of 16 k, there are actually 268 M negative examples for only 16 k positive ones. At the same time, because the sigmoid loss decomposes into a sum of per-example losses, we can perform controlled experiments to study the effect of the mini-batch composition and dis-

tribution of examples visited. We run experiments in the SigLiT setup at batch-size 16 k for 900 M steps and vary the composition of the batch by masking out (*i.e.* ignoring) enough negative examples to reach a target “positive : negative” ratio, masking in the following ways:

- **Random:** Randomly choose negative pairs to mask.
- **Hard:** Keep hardest negative pairs (highest loss).
- **Easy:** Keep easiest negatives pairs (lowest loss).
- **Hard + matching total pairs seen:** Masking examples while training for a fixed number of steps does decrease the total number of *pairs* seen during training. Hence in the *matched pairs* setting, we increase the number of training steps by the masking ratio in order to keep the number of pairs seen constant.

Figure 6 shows the effect of the various masking strategies. Randomly removing negatives to rebalance does deteriorate performance. Keeping the easiest examples does not work at all, while keeping the hardest negatives does almost

方法	图像编码器		ImageNet-1k				COCO R@1	
	ViT尺寸	# 图像块	验证集	v2	ReaL	ObjectNet	我 →T	T → 我
CLIP	B	196	68.3	61.9	-	55.3	52.4	33.1
OpenCLIP	B	196	70.2	62.3	-	56.0	59.4	42.3
EVA-CLIP	B	196	74.7	67.0	-	62.3	58.7	42.2
SigLIP	B	196	76.2	69.6	82.8	70.7	64.4	47.2
SigLIP	B	256	76.7	70.0	83.1	71.3	65.1	47.4
SigLIP	B	576	78.6	72.1	84.5	73.8	67.5	49.7
SigLIP	B	1024	79.2	73.0	84.9	74.7	67.6	50.4
CLIP	L	256	75.5	69.0	-	69.9	56.3	36.5
OpenCLIP	L	256	74.0	61.1	-	66.4	62.1	46.1
CLIPA-v2	L	256	79.7	72.8	-	71.1	64.1	46.3
EVA-CLIP	L	256	79.8	72.9	-	75.3	63.7	47.5
SigLIP	L	256	80.5	74.2	85.9	77.9	69.5	51.1
CLIP	L	576	76.6	72.0	-	70.9	57.9	37.1
CLIPA-v2	L	576	80.3	73.5	-	73.1	65.5	47.2
EVA-CLIP	L	576	80.4	73.8	-	78.4	64.1	47.9
SigLIP	L	576	82.1	75.9	87.0	81.0	70.6	52.7
OpenCLIP	G (20亿)	256	80.1	73.6	-	73.0	67.3	51.4
CLIPA-v2	H (6.3亿)	576	81.8	75.6	-	77.4	67.2	49.2
EVA-CLIP	E (50亿)	256	82.0	75.7	-	79.6	68.8	51.1
SigLIP	SO (400M)	729	83.2	77.2	87.5	82.9	70.2	52.0

表 3：与其他公开发布模型的对比。我们的 SigLIP 模型在所有现有模型（如 Open-CLIP [22] 和 CLIP [36],）之上，在零样本分类和检索任务上均取得了显著优势。与同期发布的 EVA-CLIP [43] 和 CLIPA-v2 [29], 相比，我们的 SigLIP-L 在低分辨率和高分辨率场景下均表现更优。特别值得注意的是，采用形状优化的 400M 参数 ViT [1] 架构，其性能超越了所有规模显著更大的模型。我们公开发布了我们的模型：https://github.com/google-research/big_vision。

梯度范数出现大幅尖峰，这会导致权重发生大幅度变化，可能破坏训练过程的稳定性，如图5所示。我们观察到，将 β_2 中的Adam和AdaFactor参数从默认值0.999降低至0.95（如 [20, 9]所建议），足以稳定训练。直观而言，这有助于更快地从梯度尖峰中恢复。我们选择在所有实验中设置 $\beta_2 = 0.95$ 。

4.8. Sigmoid 损失中的负样本比例

将视角从软最大化的“选择正确的类别”转换为S型函数的“评价这对样本”时，出现的一个问题是正样本对与负样本对之间的不平衡。对于批量大小 $|\mathcal{B}|$ ，该批次包含 $|\mathcal{B}|$ 个正样本对，但包含 $|\mathcal{B}|^2 - |\mathcal{B}|$ 个负样本。在16k的适中批量大小下，仅16k个正样本就对应高达2.68亿个负样本。同时，由于Sigmoid损失函数可分解为每个样本损失之和，我们能够通过受控实验来研究小批次组成及样本分布的影响。

在SigLiT设置中，我们以16k的批量大小进行9亿步的实验，并通过掩码操作（即忽略）足够多的负样本，以达到目标“正样本：负样本”比例，从而改变批次组成。掩码操作采用以下方式：

- **随机：**随机选择要遮蔽的负样本对。
- **困难：**保留最困难的负样本对（损失最高）。
- **容易：**保留最容易的负样本对（损失最低）。
- **困难 + 匹配的总训练对数量：**在固定步数训练过程中进行掩码示例，确实会减少训练期间看到的对总数。因此，在匹配对的设置中，我们根据掩码比例增加训练步数，以保持看到的对数量不变。

图6展示了各种<masking>掩码</masking>策略的效果。<Random>随机</Random>移除<negatives>负样本</negatives>以重新平衡确实会降低性能。保留最简单的样本完全无效，而保留最<Hard>困难</Hard>的<negatives>负样本</negatives>几乎能

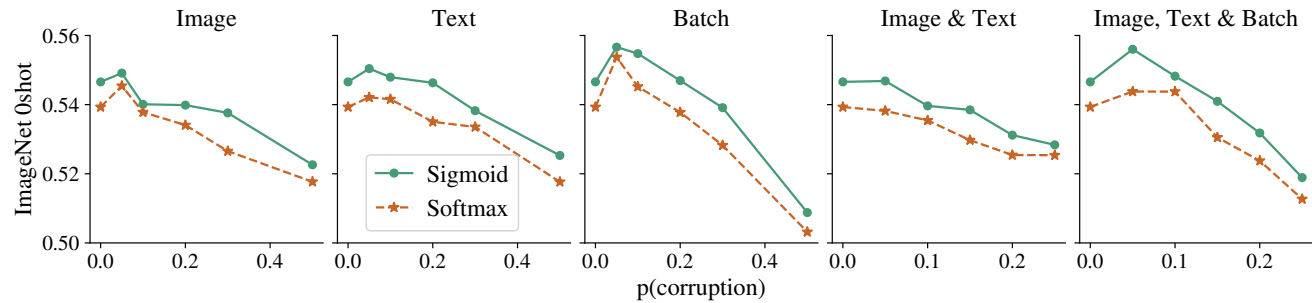


Figure 7: **Sigmoid-training increases robustness** to data noise. Titles show the type of corruption applied, and x-axes show the probability with which they are applied. With increasing corruption severity, M-scale models trained with sigmoid loss for 3.6 billion examples retain superiority over corresponding softmax baseline.

maintain the quality, indicating that, as could be expected, a lot of the learning on the negative side comes from the harder examples. This is further confirmed by the slightly increased performance of training longer on the hardest examples in order to match the total pairs seen.

We also look at the value of the learned bias at the end of training as well as the average logit value for positive and negative examples across these settings, and find the result mostly follows what one would expect: as fewer negatives are present, the bias and logits become more positive overall. Interestingly, when training with more hard negative pairs, the average logits of positive pairs stays mostly flat.

This study confirms that (1) the imbalance does not seem to be a major reason for concern, while at the same time (2) coming up with an *efficient* way of including more negative examples can be promising but is not trivial.

4.9. Bias term in sigmoid loss

We ablate the bias term in the loss function, using the Base architecture with an 8k batch size, trained for 900M examples with the SigLIP setup. Zero-shot transfer results are reported on ImageNet [14], Oxford-iiit pet [34] and Cifar100 [26]. Table 4 presents results with and without a bias term in the sigmoid loss.

Table 4: **Bias (b) and temperature (t') initialization.** Results are reported using Base architecture, 8k batch size, trained for 900M examples. Enabling the bias term b with -10 initialization improves results consistently.

b	t'	INet-0	Pet-0	C100-0
n/a	log 10	62.0	81.8	59.9
-10	log 10	63.0	82.4	61.0
-10	log 1	61.0	80.0	60.4
0	log 10	61.7	79.9	59.0
0	log 1	53.7	73.2	53.8

Enabling the bias term with a -10 initialization consistently improves performance across all tasks. This is because the bias term ensures that the training starts close to the prior, preventing dramatic over-correction in early optimization. In contrast, a randomly chosen bias term initialization, such as the 0 initialization in Table 4, fails to address the over-correction issue, leading to significantly worse results. This effect is particularly noticeable when using a small temperature t' initialization. We set the bias and temperature initialization to $b = -10$ and $t' = \log 10$ (hence $t = 10$) as the default for all experiments.

4.10. Label noise robustness

Prior works demonstrated improved robustness against label noise when using the sigmoid loss for classification models [3]. This property would be particularly useful here in the face of the famously noisy nature of popular large-scale image-text datasets. In order to study this for SigLIP, we train M/16 image models alongside an M text model at batch size 16384 for 3.6 billion seen examples. We corrupt the training data using one of the following methods:

- **Image:** With probability p , replace the image with uniform random noise.
- **Text:** With probability p , replace tokenized text with a new sequence of randomly sampled tokens, up to some (sampled) sequence length.
- **Batch alignment:** Randomly shuffle the ordering of $p\%$ of the batch.
- **Image & text:** Apply both with probability p each.
- **Image, text & batch:** Alongside (4), also shuffle fraction p of alignments.

Results from varying the likelihood of the corruption are shown in Figure 7. Models trained with sigmoid loss are increasingly robust to all kinds of added noise.

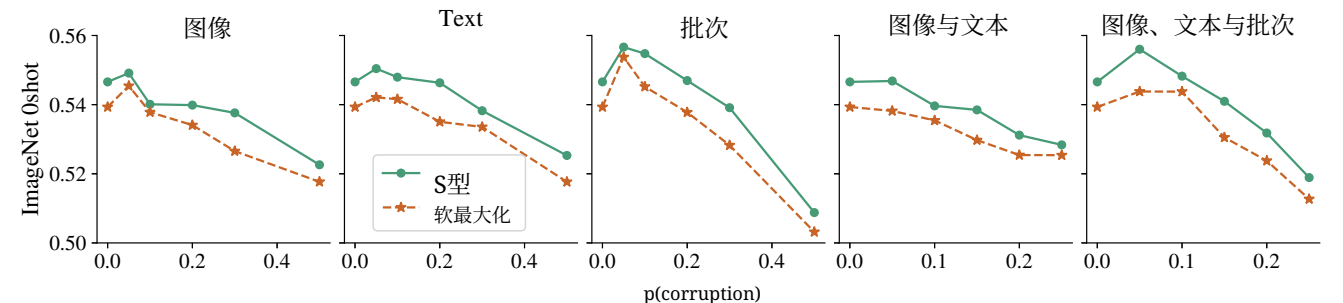


图7: **Sigmoid训练增强了对数据噪声的鲁棒性**。标题显示了所应用的损坏类型，x轴显示了其应用的频率。随着损坏严重程度的增加，使用Sigmoid损失在36亿样本上训练的M尺度模型，其性能仍优于相应的Softmax基线。

维持质量，这表明，正如预期的那样， $\langle \text{negatives} \rangle$ 负样本 $\langle \text{negatives} \rangle$ 方面的学习很大程度上来自于更 $\langle \text{Hard} \rangle$ 困难 $\langle \text{Hard} \rangle$ 的样本。这一结论通过稍微提升的性能得到了进一步证实：在最 $\langle \text{Hard} \rangle$ 困难 $\langle \text{Hard} \rangle$ 的样本上训练更长时间以匹配所见的总样本对。

我们还考察了训练结束时学习到的偏置值，以及在這些设置下正负样本的平均对数几率值，发现结果基本符合预期：随着负样本数量的减少，偏置和logit值总体上变得更正。有趣的是，当使用更多困难负样本对进行训练时，正样本对的平均logit值基本保持平稳。

本项研究证实：（1）不平衡似乎不是主要担忧的原因，同时（2）提出一种包含更多负样本的高效方法虽前景可期，但并非易事。

4.9. Sigmoid损失中的偏置项

我们消融了损失函数中的偏置项，采用8k批量大小的基础架构，在SigLIP配置下训练了9亿样本。零样本迁移结果报告于ImageNet [14], Oxford-iiit宠物 [34] 和CIFAR100 [26]上。表4展示了Sigmoid损失中带偏置项和不带偏置项的结果。

表 4: **偏置 (b) 和温度 (t') 初始化**。结果基于基础架构、8k 批量大小，训练了 9 亿个样本。启用偏置项 b 并配合 -10 初始化，可持续改进结果。

b	t'	INet-0	Pet-0	C100-0
n/a	log 10	62.0	81.8	59.9
-10	log 10	63.0	82.4	61.0
-10	log 1	61.0	80.0	60.4
0	log 10	61.7	79.9	59.0
0	log 1	53.7	73.2	53.8

启用带有 -10 初始化的偏置项能持续提升所有任务的性能。这是因为偏置项确保训练从接近先验的位置开始，防止在早期优化中出现过度修正。相比之下，随机选择的偏置项初始化（如表4中的0初始化）无法解决过度修正问题，导致结果显著变差。这一效应在使用较小的温度 t' 初始化时尤为明显。我们将偏置和温度初始化默认设置为 $b = -10$ 和 $t' = \log 10$ （即 $t = 10$ ），并作为所有实验的默认配置。

4.10. 标签噪声鲁棒性

先前的研究表明，在分类模型中使用Sigmoid损失函数可以提高对标签噪声的鲁棒性 [3]。这一特性在应对著名的大规模图像-文本数据集普遍存在的高噪声特性时尤为有用。为了研究SigLIP的这一特性，我们使用M/16图像模型和M文本模型在批量大小为16384的情况下进行训练，共训练36亿个已见样本。我们通过以下方法之一对训练数据进行破坏处理：

- **图像：**以概率 p ，将该图像替换为均匀随机噪声。
- **文本：**以概率 p ，将分词文本替换为新的随机采样词元序列，序列长度（随机采样）不超过某个值。
- **批次对齐：**随机打乱批次中 $p\%$ 的顺序。
- **图像与文本：**分别以概率 p 同时应用两者。
- **图像、文本与批次：**在 (4) 的基础上，额外随机打乱 p 部分的对齐关系。

改变损坏可能性的结果如图7所示。使用Sigmoid损失训练的模型对所有类型的添加噪声具有更强的鲁棒性。

5. Conclusion

We conducted a study on two language-image pre-training instances that used the sigmoid loss: SigLiT and SigLiP. Our results demonstrate that the sigmoid loss performs better than the softmax baseline, particularly for small train batch sizes. This loss function is also more memory efficient, which allows larger train batch sizes without requiring additional resources. We performed a thorough investigation of the batch size in contrastive learning. Surprisingly, we found that a relatively modest batch size of 32k yielded nearly optimal performance. Further studies have been performed to understand better the introduced bias term in the sigmoid loss, robustness to data noises and the impact of positive and negative pairs ratio in the sigmoid loss. We hope this work will facilitate language-image pre-training research with limited resources.

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5. 结论

我们对两个使用Sigmoid损失的语言-图像预训练实例（SigLiT和SigLiP）进行了研究。结果表明，Sigmoid损失的性能优于Softmax基线，尤其是在训练批次大小较小时。该损失函数还具有更高的内存效率，从而可以在不增加资源的情况下使用更大的训练批次大小。我们对对比学习中的批次大小进行了深入调查。令人惊讶的是，我们发现相对适中的32k批量大小就能达到近乎最优的性能。我们还进行了进一步研究，以更好地理解Sigmoid损失中引入的偏置项、对数据噪声的鲁棒性以及Sigmoid损失中正负样本对比比例的影响。我们希望这项工作能够促进在资源有限条件下的语言-图像预训练研究。

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A. More results for SigLiT

In section 4.1, we use the same precomputed embeddings for the images using a ViT-g vision model from [59]. Only resize augmentation is applied, to a fixed 288×288 resolution. We train a standard base size text tower, using the ScalingViT-Adafactor optimizer [58] with $\beta_1 = 0.9$ and $\beta_2 = 0.95$. We use 0.001 learning rate with a linear warmup schedule for the first 200M examples seen, and then the learning rate is decayed to zero with a cosine learning rate schedule. Weight decay is set to 0.0001 for all the experiments. The text is tokenized by a 32 k vocabulary sentence-piece tokenizer [27] trained on the English C4 dataset [37], and a maximum of 16 text tokens are kept. Table 8 shows results with multiple train examples seen and batch sizes, for both the sigmoid loss and the softmax loss baseline.

For training SigLiT in under a day with 4 chips (Section 4.4), we used the LION optimizer with peak learning rate 1×10^{-4} and weight decay 1×10^{-7} . The learning rate was warmed linearly to the peak in 6.5 k steps, then cosine decayed to zero for the remaining 58.5 k steps.

B. More results for SigLIP

In Table 5, we present more results for SigLIP Base with multiple train examples seen: 3 billion examples and 9 billion examples respectively.

Batch Size	3 B		9 B	
	sigmoid	softmax	sigmoid	softmax
512	51.5	47.7	-	-
1 k	57.3	53.2	-	-
2 k	62.1	59.3	-	-
4 k	65.3	63.8	68.4	66.6
8 k	68.6	66.6	70.6	69.4
16 k	-	-	72.3	71.7
32 k	69.9	69.9	73.4	72.9
98 k	69.5	69.7	73.0	73.2
307 k	-	-	71.6	72.6

Table 5: **SigLIP zeor-shot accuracy (%) on the ImageNet benchmark.** Both the sigmoid loss and the softmax loss baseline are presented. Experiments are performed on multiple train examples seen (3 B, 9 B) and train batch sizes (from 512 to 307 k). When trained for 9 B examples, the peak of the sigmoid loss comes earlier at 32 k than the peak of the softmax loss at 98 k. Together with the memory efficient advantage for the sigmoid loss, it allows one to train the best language-image model with much fewer amount of accelerators.

BS	Default	Best	Best LR	Best WD
8 k	70.1	70.1	0.001	0.0001
16 k	70.0	70.0	0.001	0.0001
32 k	68.2	69.0	0.0003	0.00003

Table 6: Default hyperparameters across different batch sizes, perform either the best or close to the best hyperparameter from a sweep. Zero-shot accuracy on ImageNet is reported. BS=batch size, LR=learning rate, WD=weight decay.

C. Robustness of SigLIP results

Hyperparameters for different batch sizes. Sigmoid loss doesn’t require tuning hyperparameters for different batch sizes. For example, in both the SigLIP and SigLiT setup, we only used default 0.001 learning rate and 0.0001 weight decay across a wide range of batch sizes (from 512 to 1024k). We further performed a sweep of 9 hyperparameters across 3 batch sizes on the from-scratch SigLIP setup for 3B seen examples: learning rate {0.0003, 0.001, 0.003} $\times$ weight decay {0.00003, 0.0001, 0.0003} $\times$ batch size {8 k, 16 k, 32 k}. We observed in Table 6 that the default LR/WD is either the best or close to the best.

Standard deviation. We repeat SigLIP training five times, using the recommended 32k batch size and 3B seen examples. We report the average and std in Table 7. The std of the five runs is very small for both sigmoid and softmax.

Alternative optimizers. We repeat the same experiment with AdamW optimizer five times and got very similar results and std as reported in Table 7. We tested a linear learning rate scheduler instead of the default cosine learning rate scheduler, it achieves 69.9% accuracy.

D. More results for mSigLIP

We present the mSigLIP Base crossmodal retrieval results on the Crossmodal-3600 dataset, across all the 36 languages in Figure 8 and Table 9.

Loss	Optimizer	Results (%)
Softmax	ViT-Adafactor	69.9 ± 0.1
Sigmoid	ViT-Adafactor	70.1 ± 0.2
Sigmoid	AdamW	70.3 ± 0.1

Table 7: Mean and standard deviation of five repeated experiments. Zero-shot accuracy on ImageNet is reported.

A. SigLiT的更多结果

在第4.1节中，我们使用来自 [59]的ViT-g视觉模型对图像采用相同的预计算嵌入。仅应用调整大小增强，将分辨率固定为 288×288 。我们使用 ScalingViT-Adafactor优化器 [58]训练一个标准尺寸的文本塔，参数设置包括 $\beta_1 = 0.9$ 和 $\beta_2 = 0.95$ 。学习率设为 0.001，采用线性预热调度，在前200M个训练样本上进行预热，之后使用余弦学习率调度将学习率衰减至零。所有实验的权重衰减均设为0.0001。文本由在英文C4数据集 [27] 上训练的32k词汇量的句子片段分词器 [37],进行分词，并最多保留16个文本令牌。表8 展示了在不同训练样本数量和批量大小下的结果，对比了Sigmoid损失函数和Softmax损失函数基线。

为了在4块芯片上一天内训练SigLiT（请参阅第4.4节），我们使用了LION优化器，其峰值学习率为 1×10^{-4} 且权重衰减为 1×10^{-7} 。学习率在6.5千步内线性升至峰值，然后在剩余的58.5千步中按余弦衰减至零。

B. SigLIP的更多结果

在表5中，我们展示了SigLIP Base在多个训练样本数量下的更多结果：分别是30亿个样本和90亿个样本。

批量大小	3 B		9 B	
	S型	软最大化	S型	软最大化
512	51.5	47.7	-	-
1 k	57.3	53.2	-	-
2 k	62.1	59.3	-	-
4 k	65.3	63.8	68.4	66.6
8 k	68.6	66.6	70.6	69.4
16 k	-	-	72.3	71.7
32 k	69.9	69.9	73.4	72.9
98 k	69.5	69.7	73.0	73.2
307 k	-	-	71.6	72.6

表5: **SigLIP在ImageNet基准测试上的零样本准确率 (%)**。同时展示了基于Sigmoid损失函数和基于Softmax损失函数的基线结果。实验在多种训练样本数量（30亿、90亿）和训练批次大小（从512到307k）下进行。当使用90亿样本训练时，基于Sigmoid损失函数的峰值在32k时出现，早于基于Softmax损失函数在98k时出现的峰值。结合Sigmoid损失函数在内存效率上的优势，这使得我们能够用更少的加速器训练出最佳的语言-图像模型。

BS	默认	Best	最佳学习率	最佳权重衰减
8 k	70.1	70.1	0.001	0.0001
16 k	70.0	70.0	0.001	0.0001
32 k	68.2	69.0	0.0003	0.00003

表6: 不同批量大小下的默认超参数，这些参数在扫描中要么表现最佳，要么接近最佳。报告了在 ImageNet上的零样本准确率。BS=批量大小，LR=学习率，WD=权重衰减。

C. SigLIP结果的鲁棒性

不同批次大小的超参数。Sigmoid损失函数无需针对不同批次大小调整超参数。例如，在SigLIP和SigLiT配置中，我们在从512到1024k的广泛批次大小范围内，仅使用了默认的0.001学习率和0.0001权重衰减。我们进一步在从头训练的SigLIP设置中，针对3个批次大小（在30亿已见样本上）对9个超参数进行了扫描：学习率 {00.0003、0.001、0.003} $\times$ 权重衰减 {00.00003、0.0001、0.0003} $\times$ 批次大小{8 16k、32k}。我们在表6 中观察到，默认的学习率/权重衰减要么是最优的，要么接近最优。

标准差。我们使用推荐的32k批次大小和30亿已见样本，重复五次SigLIP训练。我们在表7中报告了平均值和标准差。五次运行的标准差对于Sigmoid和软最大化都非常小。

替代优化器。我们使用AdamW优化器重复相同的实验五次，得到的结果和标准差与表7中报告的结果非常相似。我们测试了线性学习率调度器代替默认的余弦学习率调度器，其准确率达到69.9%。

D. mSigLIP的更多结果

我们在图8 和表9中，展示了mSigLIP Base在 Crossmodal-3600数据集上所有36种语言的跨模态检索结果。

Loss	优化器	结果 (%)
软最大化	ViT-Adafactor	69.9 ± 0.1
S型	ViT-Adafactor	70.1 ± 0.2
S型	AdamW	70.3 ± 0.1

表 7: 五次重复实验的均值与标准差。报告为在 ImageNet上的零样本准确率。

Batch Size	450 M		900 M		3 B		18 B	
	sigmoid	softmax	sigmoid	softmax	sigmoid	softmax	sigmoid	softmax
512	72.5	69.5	75.0	72.8	77.2	74.6	-	-
1 k	75.5	73.6	77.2	76.0	79.6	77.9	-	-
2 k	77.1	76.3	79.3	78.1	81.3	80.1	82.2	81.2
4 k	79.2	78.3	80.8	79.8	82.4	81.2	83.0	82.0
8 k	80.8	79.7	82.0	81.0	83.1	82.6	83.6	83.1
16 k	81.2	81.2	82.7	82.1	83.8	83.5	84.2	84.1
32 k	81.9	81.4	83.1	82.7	84.2	84.0	84.6	84.4
64 k	81.6	81.6	83.0	82.8	84.3	84.1	84.7	84.4
128 k	80.5	80.0	83.1	83.2	84.2	84.4	84.7	84.6
256 k	72.8	72.2	82.1	81.7	84.3	84.2	84.7	84.6
1024 k	-	-	-	-	-	-	84.7	-

Table 8: **SigLiT zero-shot accuracy (%) on the ImageNet benchmark.** Both the sigmoid loss and the softmax loss baseline are presented. Extensive experiments are performed on multiple train examples seen (450 M, 900 M, 3 B, 18 B) and train batch sizes (from 512 to 1 M).

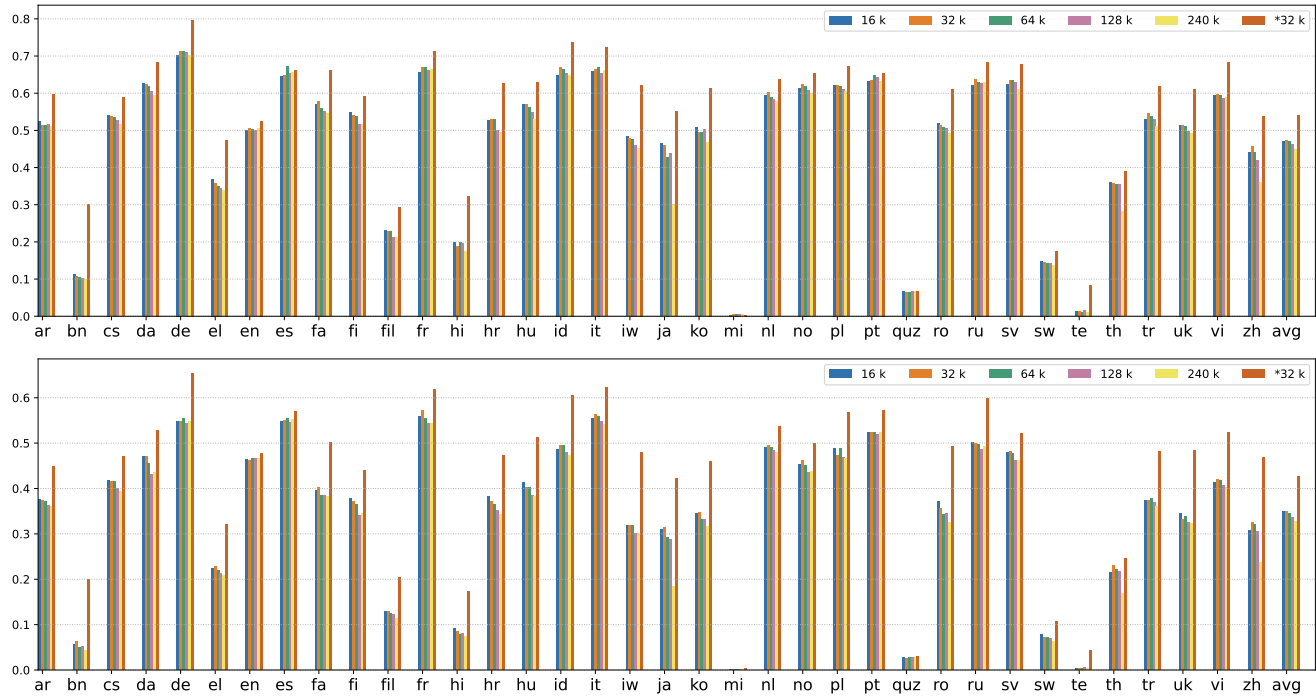


Figure 8: **Image-to-text and text-to-image zero-shot retrieval recall@1 results on all 36 languages of Crossmodal-3600.** Top: Image to text. Bottom: text to image. Colors are batch sizes. *32 k represents the scaled up results as described in Section 4.6.

E. Label noise experiments

All models had an M/16 image tower and a M text tower. They were trained from random initialisation for 3.6B examples seen, with a batch size of 16384. A cosine learning rate schedule was used, with an initial linear warmup for 10% of steps up to a peak learning rate of 0.001.

批量大小	450 M		900 M		3 B		18 B	
	S型	软最大化	S型	软最大化	S型	软最大化	S型	软最大化
512	72.5	69.5	75.0	72.8	77.2	74.6	-	-
1 k	75.5	73.6	77.2	76.0	79.6	77.9	-	-
2 k	77.1	76.3	79.3	78.1	81.3	80.1	82.2	81.2
4 k	79.2	78.3	80.8	79.8	82.4	81.2	83.0	82.0
8 k	80.8	79.7	82.0	81.0	83.1	82.6	83.6	83.1
16 k	81.2	81.2	82.7	82.1	83.8	83.5	84.2	84.1
32 k	81.9	81.4	83.1	82.7	84.2	84.0	84.6	84.4
64 k	81.6	81.6	83.0	82.8	84.3	84.1	84.7	84.4
128k	80.5	80.0	83.1	83.2	84.2	84.4	84.7	84.6
256 k	72.8	72.2	82.1	81.7	84.3	84.2	84.7	84.6
1024 k	-	-	-	-	-	-	84.7	-

表8: **SigLiT在ImageNet基准测试上的零样本准确率 (%)**。同时展示了Sigmoid损失函数和Softmax损失函数基线。我们在多种训练样本数量（4.5亿、9亿、30亿、180亿）和训练批次大小（从512到100万）上进行了大量实验。

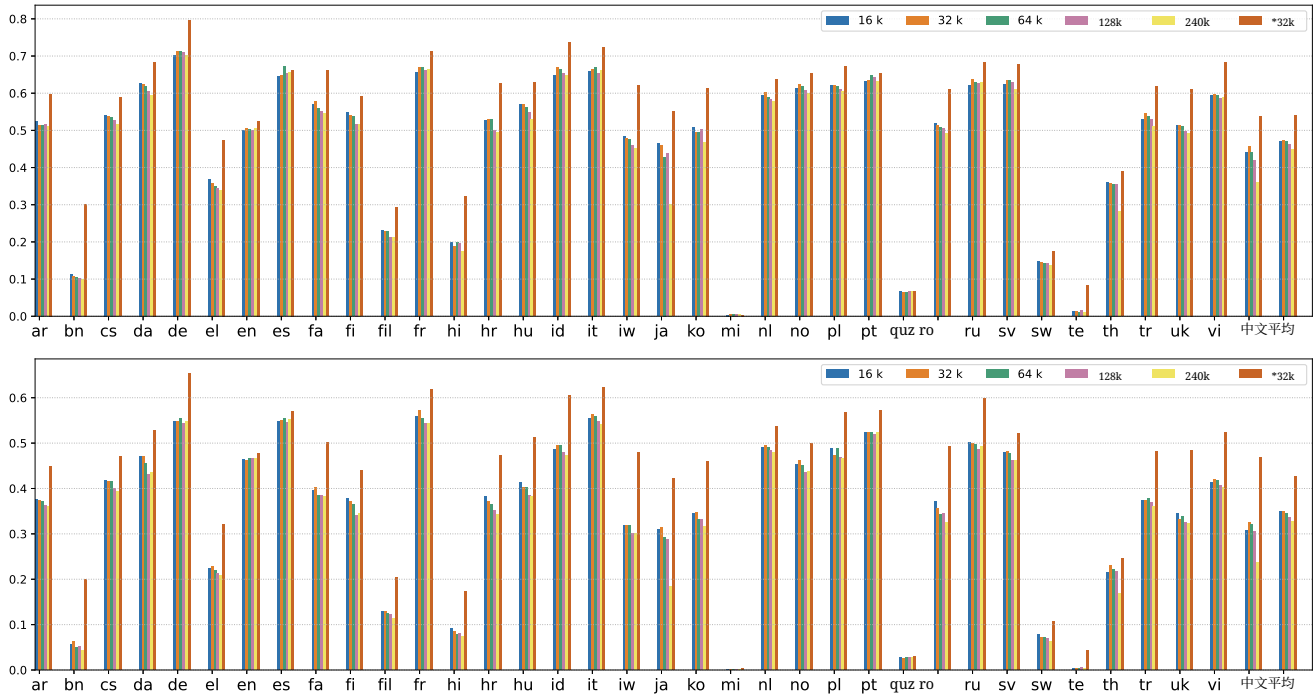


图8: **Crossmodal-3600所有36种语言上的图像到文本和文本到图像零样本检索召回率@1结果**。上方：图像到文本。下方：文本到图像。颜色代表批次大小。*32k代表第4.6节中描述的放大结果。

E. 标签噪声实验

所有模型均采用M/16图像塔和M文本塔。它们从随机初始化开始训练，处理了36亿个样本，批量大小为16384。使用余弦学习率调度，并采用初始线性预热，预热步数占总步数的10%，峰值学习率为0.001。

Lang.	Image-to-text						Text-to-image					
	16 k	32 k	64 k	128 k	240 k	*32 k	16 k	32 k	64 k	128 k	240 k	*32 k
ar	52.4	51.3	51.5	51.5	51.1	59.7	37.6	37.4	37.1	36.3	36.0	44.9
bn	11.4	10.8	10.4	10.3	9.9	30.1	5.5	6.2	4.9	5.1	4.4	20.0
cs	54.1	53.7	53.7	52.8	51.8	58.9	41.8	41.6	41.5	39.9	39.4	47.0
da	62.7	62.4	62.0	60.4	59.3	68.4	47.0	47.0	45.6	43.0	43.5	52.9
de	70.3	71.4	71.2	71.1	70.2	79.7	54.7	54.8	55.4	54.3	54.7	65.3
el	36.9	35.8	35.1	34.5	33.8	47.4	22.4	22.8	22.0	21.3	20.8	32.2
en	50.1	50.5	50.2	49.9	50.7	52.5	46.5	46.2	46.5	46.6	46.6	47.6
es	64.7	64.9	67.2	65.3	65.6	66.3	54.8	55.0	55.5	54.5	55.2	57.0
fa	57.0	57.8	56.1	55.3	54.6	66.2	39.6	40.2	38.4	38.4	38.3	50.0
fi	54.9	54.1	53.8	51.7	51.7	59.1	37.7	37.1	36.4	34.0	34.5	44.0
fil	23.2	22.8	22.9	21.4	21.2	29.2	12.8	12.9	12.4	12.2	11.3	20.4
fr	65.7	66.9	67.0	66.1	66.5	71.2	55.9	57.1	55.5	54.4	54.3	61.8
hi	19.9	18.8	19.9	19.5	17.4	32.2	9.1	8.5	7.9	8.1	7.3	17.3
hr	52.7	53.0	53.0	49.9	49.6	62.6	38.2	37.1	36.4	35.2	34.3	47.2
hu	57.0	57.1	56.3	54.8	53.0	62.9	41.4	40.2	40.2	38.6	38.2	51.2
id	64.8	67.1	66.6	65.4	64.7	73.7	48.5	49.4	49.5	47.8	47.3	60.5
it	65.9	66.4	67.1	65.2	66.1	72.3	55.5	56.4	55.8	54.8	54.1	62.3
iw	48.4	47.9	47.7	46.1	45.2	62.2	31.8	31.8	31.9	30.1	30.1	48.0
ja	46.4	45.9	42.9	43.7	30.2	55.1	31.0	31.3	29.2	28.9	18.5	42.3
ko	50.8	49.5	49.4	50.2	46.8	61.4	34.4	34.7	33.2	33.1	31.5	45.9
mi	0.4	0.4	0.6	0.6	0.4	0.3	0.2	0.2	0.2	0.2	0.2	0.3
nl	59.6	60.4	58.9	58.3	57.9	63.6	48.9	49.5	48.9	48.4	47.9	53.6
no	61.4	62.4	62.0	60.9	59.9	65.3	45.3	46.2	45.0	43.5	43.7	50.0
pl	62.2	62.0	62.0	61.1	60.5	67.1	48.8	47.4	48.7	46.8	46.7	56.7
pt	63.1	63.6	64.9	64.3	63.2	65.4	52.4	52.3	52.3	51.9	52.4	57.3
quz	6.8	6.4	6.4	6.6	6.7	6.8	2.7	2.6	2.7	2.7	2.8	2.9
ro	52.1	51.4	51.0	50.6	49.3	61.0	37.2	35.6	34.3	34.5	32.5	49.3
ru	62.2	63.6	63.1	62.7	63.1	68.4	50.1	49.9	49.7	48.6	49.3	59.9
sv	62.3	63.5	63.5	63.1	61.2	67.7	47.9	48.2	47.6	46.2	46.2	52.0
sw	14.8	14.4	14.3	14.2	13.8	17.4	7.8	7.2	7.1	6.9	6.3	10.7
te	1.2	1.2	1.2	1.7	1.1	8.4	0.4	0.3	0.3	0.5	0.3	4.3
th	36.1	35.8	35.6	35.6	28.3	39.0	21.6	23.1	22.2	21.6	16.8	24.6
tr	53.1	54.5	53.7	52.9	51.2	62.0	37.3	37.4	37.8	37.0	36.1	48.1
uk	51.4	51.5	51.2	49.9	49.2	61.2	34.5	33.2	33.8	32.5	32.4	48.3
vi	59.6	59.8	59.5	58.5	58.8	68.4	41.4	41.9	41.9	40.6	40.3	52.3
zh	44.1	45.7	44.1	41.9	36.1	53.9	30.7	32.5	32.0	30.6	23.7	46.8
avg	47.2	47.4	47.1	46.3	45.0	54.1	34.8	34.9	34.4	33.6	32.7	42.6

Table 9: **Image-to-text (text retrieval) and text-to-image (image retrieval) zero-shot recall@1 results on all 36 languages of Crossmodal-3600**, with mSigLIP models trained at different batch sizes for 30 B total examples seen. *32k represents the scaled up results as described in Section 4.6.

语言	图像到文本						文本到图像					
	16 k	32 k	64 k	128k	240k	*32k	16 k	32 k	64 k	128k	240k	*32k
ar	52.4	51.3	51.5	51.5	51.1	59.7	37.6	37.4	37.1	36.3	36.0	44.9
bn	11.4	10.8	10.4	10.3	9.9	30.1	5.5	6.2	4.9	5.1	4.4	20.0
cs	54.1	53.7	53.7	52.8	51.8	58.9	41.8	41.6	41.5	39.9	39.4	47.0
da	62.7	62.4	62.0	60.4	59.3	68.4	47.0	47.0	45.6	43.0	43.5	52.9
de	70.3	71.4	71.2	71.1	70.2	79.7	54.7	54.8	55.4	54.3	54.7	65.3
el	36.9	35.8	35.1	34.5	33.8	47.4	22.4	22.8	22.0	21.3	20.8	32.2
en	50.1	50.5	50.2	49.9	50.7	52.5	46.5	46.2	46.5	46.6	46.6	47.6
es	64.7	64.9	67.2	65.3	65.6	66.3	54.8	55.0	55.5	54.5	55.2	57.0
fa	57.0	57.8	56.1	55.3	54.6	66.2	39.6	40.2	38.4	38.4	38.3	50.0
fi	54.9	54.1	53.8	51.7	51.7	59.1	37.7	37.1	36.4	34.0	34.5	44.0
fil	23.2	22.8	22.9	21.4	21.2	29.2	12.8	12.9	12.4	12.2	11.3	20.4
fr	65.7	66.9	67.0	66.1	66.5	71.2	55.9	57.1	55.5	54.4	54.3	61.8
hi	19.9	18.8	19.9	19.5	17.4	32.2	9.1	8.5	7.9	8.1	7.3	17.3
hr	52.7	53.0	53.0	49.9	49.6	62.6	38.2	37.1	36.4	35.2	34.3	47.2
hu	57.0	57.1	56.3	54.8	53.0	62.9	41.4	40.2	40.2	38.6	38.2	51.2
id	64.8	67.1	66.6	65.4	64.7	73.7	48.5	49.4	49.5	47.8	47.3	60.5
it	65.9	66.4	67.1	65.2	66.1	72.3	55.5	56.4	55.8	54.8	54.1	62.3
iw	48.4	47.9	47.7	46.1	45.2	62.2	31.8	31.8	31.9	30.1	30.1	48.0
ja	46.4	45.9	42.9	43.7	30.2	55.1	31.0	31.3	29.2	28.9	18.5	42.3
ko	50.8	49.5	49.4	50.2	46.8	61.4	34.4	34.7	33.2	33.1	31.5	45.9
mi	0.4	0.4	0.6	0.6	0.4	0.3	0.2	0.2	0.2	0.2	0.2	0.3
nl	59.6	60.4	58.9	58.3	57.9	63.6	48.9	49.5	48.9	48.4	47.9	53.6
no	61.4	62.4	62.0	60.9	59.9	65.3	45.3	46.2	45.0	43.5	43.7	50.0
pl	62.2	62.0	62.0	61.1	60.5	67.1	48.8	47.4	48.7	46.8	46.7	56.7
pt	63.1	63.6	64.9	64.3	63.2	65.4	52.4	52.3	52.3	51.9	52.4	57.3
quz	6.8	6.4	6.4	6.6	6.7	6.8	2.7	2.6	2.7	2.7	2.8	2.9
ro	52.1	51.4	51.0	50.6	49.3	61.0	37.2	35.6	34.3	34.5	32.5	49.3
ru	62.2	63.6	63.1	62.7	63.1	68.4	50.1	49.9	49.7	48.6	49.3	59.9
sv	62.3	63.5	63.5	63.1	61.2	67.7	47.9	48.2	47.6	46.2	46.2	52.0
sw	14.8	14.4	14.3	14.2	13.8	17.4	7.8	7.2	7.1	6.9	6.3	10.7
te	1.2	1.2	1.2	1.7	1.1	8.4	0.4	0.3	0.3	0.5	0.3	4.3
th	36.1	35.8	35.6	35.6	28.3	39.0	21.6	23.1	22.2	21.6	16.8	24.6
tr	53.1	54.5	53.7	52.9	51.2	62.0	37.3	37.4	37.8	37.0	36.1	48.1
uk	51.4	51.5	51.2	49.9	49.2	61.2	34.5	33.2	33.8	32.5	32.4	48.3
vi	59.6	59.8	59.5	58.5	58.8	68.4	41.4	41.9	41.9	40.6	40.3	52.3
zh	44.1	45.7	44.1	41.9	36.1	53.9	30.7	32.5	32.0	30.6	23.7	46.8
avg	47.2	47.4	47.1	46.3	45.0	54.1	34.8	34.9	34.4	33.6	32.7	42.6

表9：基于mSigLIP模型，在不同的批量大小下训练，总样本数为300亿时，在Crossmodal-3600全部36种语言上的图像到文本（文本检索）和文本到图像（图像检索）的零样本召回率@1结果，*32k表示第4.6节所述的放大结果。

F. Model Card

We provide a description of our models following [32].

- **Model Architecture:** The model is trained using the contrastive pre-training technique with sigmoid loss as described in this paper. This contrastive model contains two encoders, i.e. vision transformer encoder [17] and language transformer encoder [47]. The vision and language encoders always have the same size, one of ViT-B, ViT-L and SoViT-400M [1].
- **Inputs:** The vision encoder takes an image ($224 \times 224 \times 3$, $256 \times 256 \times 3$, $384 \times 384 \times 3$, $512 \times 512 \times 3$) as input. The text encoder takes a tokenized text [38, 54] cropped to the first 64 tokens as input.
- **Outputs:** The vision and text encoders both output a d dimensional feature vector, where d is 768, 1024 and 1152 for ViT-B, ViT-L and SoViT-400M, respectively.
- **Intended Use:** The models are designed for multi-modal research purposes. The models can be used for zero-shot image classification and zero-shot image-text retrieval by comparing both feature vectors. We provide both en-only and i18n-trained models to encourage research on the impact of this choice.
- **Training Data:** The contrastive model is pre-trained from-scratch using the WebLI [13] dataset. SigLIP models are pre-trained on a WebLI subset filtered to contain mostly English. mSigLIP models are pre-trained on the WebLI dataset without language filters.
- **Evaluation Data:** Zero-shot classification is performed on ImageNet [14], ImageNet v2 [39], ImageNet Real [3], and ObjectNet [2]. Zero-shot retrieval is performed on COCO [11] and the multilingual XM3600 dataset [44].
- **Hardware & Software:** The models are developed in the `big_vision` codebase [5, 4] and trained on Google Cloud TPUs.

F. 模型卡

我们对模型的描述如下 [32]。

- **模型架构：**该模型采用本文所述的Sigmoid损失对比预训练技术进行训练。该对比模型包含两个编码器，即视觉变换器编码器 [17]和语言Transformer编码器 [47]。视觉编码器和语言编码器的大小始终相同，包括ViT-B、ViT-L和SoViT-400M [1]。
- **输入：**视觉编码器以图像（ $224 \times 224 \times 3$ ， $256 \times 256 \times 3$ ， $384 \times 384 \times 3$ ， $512 \times 512 \times 3$ ）作为输入。文本编码器以裁剪至前64个标记的分词文本 [38, 54]作为输入。
- **输出：**视觉编码器和文本编码器均输出 d 维特征向量，其中对于ViT-B、ViT-L和SoViT-400M， d 分别为768、1024和1152。
- **预期用途：**这些模型专为多模态研究目的而设计。模型可用于零样本图像分类和零样本图像-文本检索，通过比较两者的特征向量实现。我们提供了纯英文和多语言训练的模型，以鼓励研究这一选择所带来的影响。
- **训练数据：**该对比模型使用WebLI数据集进行从头训练。SigLIP模型在WebLI数据集的子集上进行预训练，该子集经过过滤，主要包含英文内容。mSigLIP模型则在未经过语言过滤的WebLI数据集上进行预训练。
- **评估数据：**零样本分类在ImageNet [14]，ImageNet v2 [39]，ImageNet ReaL [3]，和 ObjectNet [2]上进行。零样本检索在COCO [11]以及多语言XM3600数据集 [44]上进行。
- **硬件与软件：**这些模型在big vision代码库 [5, 4] 中开发，并在谷歌云TPU上进行训练。